

Introduction

Vance Russell, Christopher Sacco

About

The [Forest Business Alliance](#) provides technical assistance, delivers workshops, and facilitates a peer-learning network to increase local and regional capacity for applicants to CAL FIRE's Business and Workforce Development Program.

Funding

Funding for the Forest Business Alliance is provided by the [CAL FIRE Business and Workforce Development Grant Program](#). Past funders include the [Yuba Water Agency](#) and USDA Forest Service.



Audience

Businesses, Tribal organizations, nonprofits, and other entities are creating sustainable wood product facilities and businesses. The guide provides key insights and advice on accessing funds. More information and templates for the CAL FIRE Business and Workforce Development and USDA Forest Service Wood Innovation Grants can be found at the [Forest Business Alliance](#) website. Feedback is welcome. Contact us directly at forestbusinessalliance@gmail.com.

Contents

We will add new chapters as they become ready. Please check back regularly!

1. **Introduction.** This chapter.
2. **Program Design.** How to create a program or strategy for your organization/business.
3. **Proposals.** A startup guide for project design and writing proposals.
4. **Monitoring.** Monitoring and evaluation of your program and projects. Testing assumptions inherent in your theory of change.
5. **eCivis.** Managing your application and project using eCivis.
6. **Compliance Pathways.** Overview of the California Environmental Quality Act and its application to forest-based business/organization projects.
7. **Resilience Gap.** An analysis by New Leaf Climate Partners examining fuel reduction needs to decrease risks of severe wildfire for private forest owners in low-income communities in California.
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12. **Cases.** Case studies of successful business and workforce development in California's forest health sector. The first case featured the Washoe Tribe's Elder, Share & Sale Firewood Program followed by Plumas Wood Fiber and Forestree Collective. More to come soon.
13. **Grant Closeout.** How to leverage and learn from closing out your grant.

Program Design

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Key Concepts

Key concepts in this document include:

- **Mission alignment.** Explicitly designing programs before seeking to fund them helps businesses and organizations stay true to their missions. Without a well-designed program aimed at achieving long-term goals, the ever-present need to raise money can push you to pursue funding opportunities that are only peripherally related to your purpose. This “tail wagging the dog” scenario is less likely to occur when your team has gone through a thoughtful design process and has a document to refer back to when deciding whether or not to pursue new opportunities.
- **Program vs. project.** A project is a set of activities a team implements to achieve a specific objective. A program is a set of related projects, each with its own objectives, that work in concert to achieve a wider goal(s). A team guided by a well-designed program will carefully select projects based on their potential to contribute to institutional goals. Also, thoughtfully choosing projects can allow organizations to maximize finite resources—people, equipment, supplies, and money—thus increasing productivity and operational efficiency.
- **Acknowledge assumptions.** All programs and projects have inherent assumptions in how they are implemented. Acknowledge these in your theory of change and periodically review whether they are correct. In other words, implement the tasks and interventions to solve the problem or create a solution to a problem that leads to the (assumed) intended outcomes.
- **Iterate and scale.** Start small, experiment, and scale your successes Bunch (1982). Scaling successes is not just about expanding; it involves planning and cost estimates across a wider scope/region, feasible analytical methods, and strategy to theory of change linkages Salafsky *et al.* (2021).

Background

The Forest Business Alliance (FBA) created this chapter to help grant seekers, program/project managers, and business owners design projects for submission to CAL FIRE’s Wood Products and Bioenergy Business and Workforce Development Grant [Program](#). These guidance documents may also be useful for the USDA Forest Service’s Wood Innovation [Program](#) and other grant programs.

However, while providing technical assistance to businesses and nonprofits, we realized there was too much emphasis on creating projects to respond to proposal requests. Many organizations were developing projects without considering how they fit into a program or the broader organizational purpose, or, at worst, chasing funds without a plan or organizational priorities. Therefore, the guide emphasizes taking a step back from project planning to consider how it may fit into programmatic priorities and how you may need to retrofit strategic-level priorities in your organization.

Program vs. Project

Program = A long-term collection of interrelated projects designed to achieve strategic goals and supporting objectives.

Project = A time-bound endeavor undertaken to produce a specific output or deliverable. A suite of projects should contribute to program goals.

A project may be analogous to building a house, whereas a program, depending on its complexity and scale, could be creating a neighborhood or city.

Competitive grant proposals are built around well-designed projects rooted in and contribute to thoughtfully planned programs. Projects that are well-designed are most effectively identified through proactive program planning. Program planning moves from the big picture (problems, needs, overarching goals, and desired

outcomes) to an action-oriented project scale (what will you do to achieve the goals and outcomes, how long will that take, and how much will it cost).

This guidance document and the accompanying worksheets are intended to be used step-by-step to move you through the following design process:

1. Situation assessment
2. Theory of Change
3. Goals & Objectives
4. Monitoring (see monitoring chapter for more detail)
5. Budget
6. Implementation

Please keep in mind that program design is iterative and dynamic. As you add detail and progress, the information you input during prior steps may need to be revised.

We used numerous publicly available sources to create the project design and monitoring templates. We adapted information to create resources suited to the CAL FIRE Business and Workforce Development Program grant program. The Open Standards for the Practice of Conservation, developed by the Conservation Measures Partnership (CMP, 2025), and ProPack I: Project Design Guidance for Project and Program Managers, developed by Catholic Relief Services (CRS, 2015), are among the key sources we relied upon to develop this chapter.

Templates

The templates and information provided are for reference only. While FBA has made every effort to provide up-to-date and correct information, we make no representations concerning the contents.

Assessment

Program planning typically begins with a participatory assessment of your business and organization and with the board, stakeholders, and clients. Conducting a thorough situation or problem analysis is central to your assessment. Your organization or business should review available evidence to complete your analysis, identifying the direct threats, core problems, and indirect threats, also known as drivers and causes. Evidence can be in many formats, for example [Conservation Evidence](#), local studies or research on forest health, interviews with experts who possess historical knowledge and experience, or proven projects, pilots, or technologies with data to back up their outcomes.

A new or revised assessment should be completed when preparing grant proposals, designing new programs, or when several years have passed since a previous assessment.

Important

Situation Assessment = A model of the system in which conservation programs are taking place and trying to effect change Salafsky *et al.* (2021).

We recommend you review pages 13 through 24 of the Open Standards CMP (2025), which provides an excellent primer on assessment.

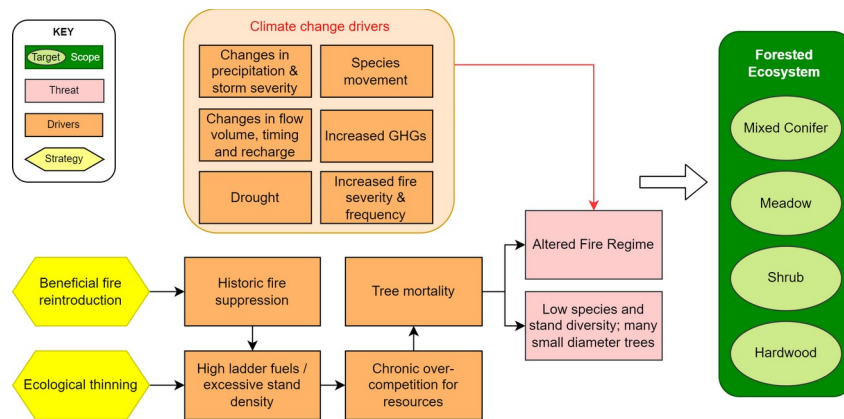


Figure 1: Simplified forest health situation model.

A situation model is a tool that visually portrays the relationships among the various factors in your situation analysis (Figure 1). When presented visually, this analysis is more powerful. Both tools can explore and ultimately illustrate a program and its context. The model should illustrate the main cause-and-effect relationships and include the most important details, yet be as simple as possible. To the degree that it is feasible and useful, you should identify the actors behind key factors and discuss assumptions between the relationship factors. For example, in Figure 1 arrows are essentially assumptions about the relationships between boxes in the assessment, e.g., tree mortality leads to increased fuel in a stand and alters the fire regime, especially in the absence of fire.

Although Figure 2 depicts a conservation-focused model, a business can easily adapt it to suit its business model. The drivers and threats to healthy forests will remain the same in a forested system, but the strategy, scope, and targets may change. For instance, a wood products business may be focused on utilizing the wood removed by thinning projects and possibly increasing workforce capacity. Adapting the situation model to a forest business may look like Figure 2.

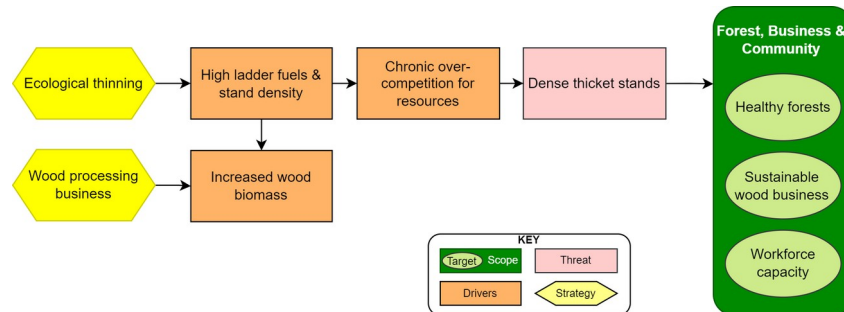


Figure 2: Wood products business situation model.

Design teams frequently use assessment findings to construct problem trees (Figure 3). These causal diagrams help you identify and study core problems by identifying their immediate and underlying causes and their negative effects. This analysis, in turn, allows you to identify solutions and desired outcomes that will address the problems rather than their symptoms. A problem tree is also useful for understanding the linkages between your program actions and desired outcomes.

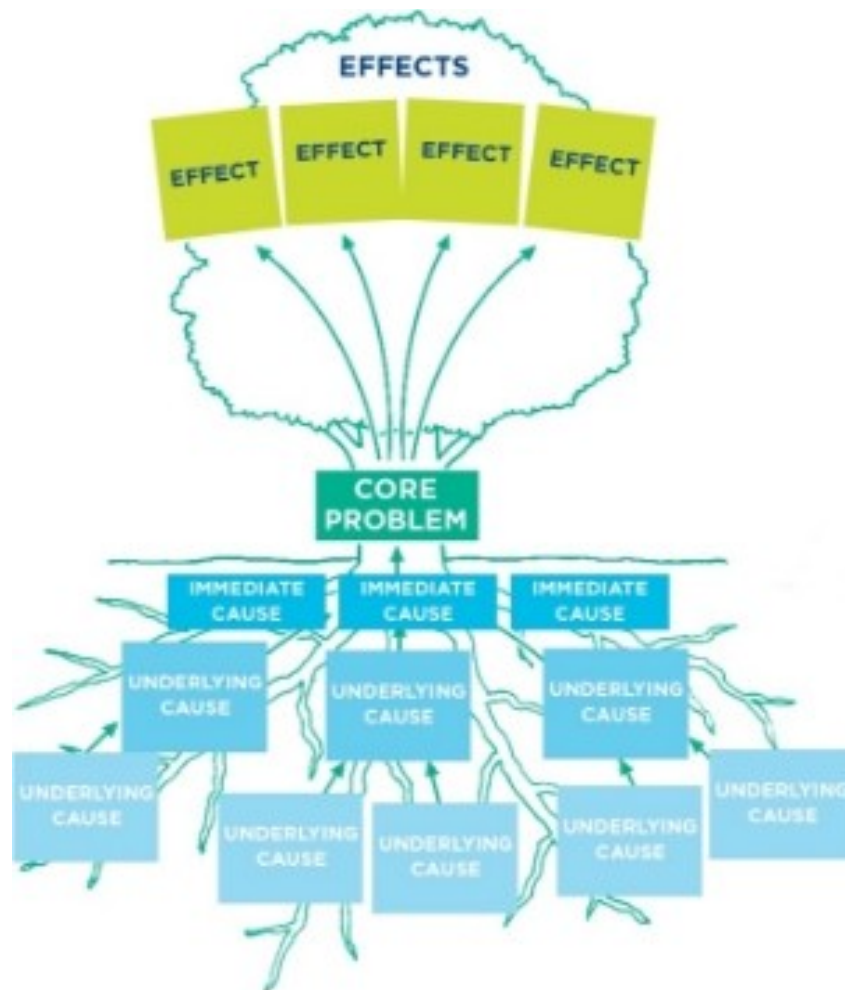


Figure 3: Problem tree.

Creating a Problem Tree

Problem tree analysis is ideally done with stakeholders at a participatory workshop. Participants write problems, immediate causes, root causes, and effects on sticky notes or note cards. Each card should contain only one item. This allows participants to move the cards around and discuss where they best fit within the problem tree.

Once you have created your situation model or problem trees, review the logic by asking these questions:

- Does each cause-effect link make sense? Is each link plausible? Why or why not?
 - How well have the causes connected to the roots? Are there any unidentified root causes?
 - What appears to be the relative contribution of each causal stream to the problem/threat? Do some causes appear more than once? Why is this?

Theory of Change

A Theory of Change bridges the problem analysis visualized in the problem tree(s) and the proposed responses reflected in the project's results framework or ProFrame Figure 4. The theory of change clarifies why you believe

the selected project strategies will work in the project context and justifies and checks the logic and feasibility of the change hypothesis. A theory of change can be expressed in narrative or diagrammatic form.

Theory of Change

A narrative Theory of Change is a concise, explicit explanation of:

“IF we implement strategies 1, 2, and 3 and our assumptions hold true THEN, we will achieve the following interim results and outcomes BECAUSE of a defensible/cited evidence base”

With this structure, the theory of change clarifies how (if-then) and why (because) the project team expects or assumes that certain actions will produce desired changes for individuals, groups, communities, or institutions in the environment where the project will be implemented (Figure 4).

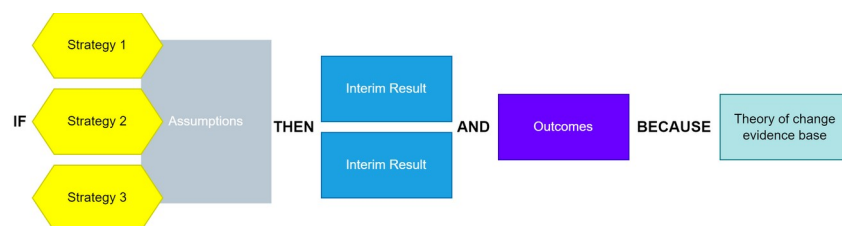


Figure 4: Simplified theory of change.

The theory of change is not simply a narrative description of the results framework because the results framework only reflects the elements (the “ifs” and “then” parts) that the project will deliver, whereas the theory of change also states those actions or contributions critical to the project success but which you expect other actors to deliver. In other words, the theory of change reflects both the results framework and the project’s hypotheses and assumptions. Remember that the theory of change can be developed for different levels of the objectives hierarchy. A “high-level” theory of change articulates how achieving the project’s objectives will contribute to longer-term, broader, lasting change (project’s goal). You should minimally create a theory of change at this level. However, it may be necessary to create links from all project strategies to outcomes.

Results Chains

Instead of a narrative theory of change, you may use a results chain, a visual diagram of a theory of change (Figure 5). The results chain shows the strategies identified in the situation model and connects them to interim results and project outcomes.



Figure 5: Results chains are a visual diagram of a theory of change showing anticipated outcomes CMP (2025).

A more specific example of workforce capacity building is shown in Figure 6. Once you have finished developing all project results chains, you can write your goals and objectives.

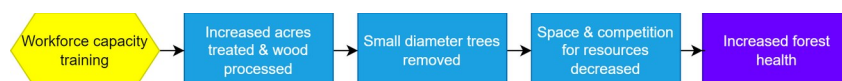


Figure 6: A simplified workforce capacity results chain.

Goals & Objectives

Goals and objectives are often reversed, conflated, or used interchangeably by organizations, businesses, and agencies. Goals and objectives are distinct, important planning terms that identify programmatic success when conceived and well-written, e.g., how you want the world to change. Below, we provide guidance on creating statements for each level. Review CAL FIRE's required metrics before you develop your goals and objective statements. Reviewing and understanding them is necessary for creating goals and objectives that align with CAL FIRE's goals.

Wood Business Success Pathways

Goals and objectives are critical for program design, maintaining focus, and measuring success over time. Some key factors for successful wood product and forest health businesses and organization include

1. **Business plan.** Develop a clear business plan with a vision, achievable objectives, scalable expansion, and brief, but accurate, finance plan.
2. **Networking.** Network across professional in the field to see their approaches, how they've solved problems, and develop and possible partnerships.
3. **Compliance.** Start and CEQA/NEPA compliance early and budget properly for the documents and permits you need to get that first spade in the ground.
4. **Feedstock agreements.** Make sure to secure long-term feedstock agreements from the landowner where your wood is sourced. It can be exceptionally difficult to arrange terms longer than three years with public agencies, so the earlier you start, the better. For more detail on this topic and potential solutions, see Russell (2025).
5. **Funding.** This is the big one, and the most challenging. However, if you organize your plans, your pitch deck, and your business idea is good and fills a key need, the funding will come. Don't confuse funding and finance, they're two very different things, but you need to do both. There are many options for collaborative finance in the forestry and water worlds (Russell & Odefey, 2024).

Goals are the high-level, long-term results and impacts an organization or business seeks to achieve. Derived from the scope of your situation assessment, they represent the desired status of the specific targets identified. Like objectives, goals are specific, measurable, and time-bound but typically over a much longer period, e.g., decades rather than months or years. Unlike objectives, the project's goal statement is usually general and abstract, describing a desired state beyond the project's life. For this reason, a goal will not have an associated indicator. Program goals are the nexus for your organization's projects.

Goal Example

By 2035, forest health will be significantly improved in California's Central Coast region through management techniques such as thinning and prescribed fire.

Objectives represent the higher-order changes in systems, communities, or organizations that can reasonably be achieved during a project's time frame and provide the building blocks to achieve goals. Well-written objectives are SMART = specific, measurable, achievable, results-oriented, and time-bound and are tied to the intermediate results and outcomes in a theory of change or results chain. In many projects, objectives reflect the benefits expected to occur by the end of the project as a result of behavioral changes at the interim results level. Goals are how you want the world to be, and objectives describe what you will do to reach those goals.

Objective Example

By September 2025, 300 future thinning and prescribed fire crew members from the Central Coast Region will be trained and red-carded.

Interim results state the expected changes in identifiable behaviors by participants or in identifiable approaches by interventions, systems, policies, or institutions as a result of what was gained (outputs) through project actions (activities). Progress at this level is a necessary precondition for achieving the objectives. Write intermediate results as complete sentences, as if already achieved. Put the targeted primary beneficiary group(s) whose behavior is expected to change as the sentence's subject. Interim results may focus on demonstrable evidence of behavior change, such as adoption, uptake, or coverage.

Interim Results Example

Training organizations will coordinate to identify 1,000 potential crew members from disadvantaged communities of the Central Coast to participate in comprehensive thinning and prescribed fire practical courses by 2023.

Next Steps

1. **Monitoring Plan.** Once the situation assessment, theory of change, and programmatic goals/objectives are completed, it's time to develop a monitoring plan. We will cover this in more depth in Chapter 3.
2. **Project Plans.** Project planning follows and can be driven and prioritized by your program strategy. See the project planning chapter for details.
3. **Analysis and Learning.** As your business or organization matures and your project experiences yield successes and data you track, it is time to analyze and learn from the results periodically. We'll cover this in Chapter 3. However, reflective organizations/businesses learn the most from their experience, including mistakes, and use this learning to build a better organization. Don't short-shrift this component! Set aside organizational time to complete it through regular analysis, learning, and writing workshops or strategic planning retreats.

Resources

- The [Open Standards for the Practice of Conservation](#), developed by the Conservation Measures Partnership CMP (2025) is a definitive resource for program and project design.
- [ProPack I: Project Design Guidance for Project and Program Managers](#), developed by Catholic Relief Services CRS (2015) is a useful resource.
- Business planning differs from organizational planning and typically focuses on developing a strategic business plan and pitch deck. See the [Forest Business Alliance](#) templates page for a business plan template. For a forest product business plan, see [Klamath River Forest Products](#) business plan.
- [Miradi](#) is designed to help project proponents move through and plan the various stages of a program and projects and is widely adaptable.
- [ArcGIS StoryMaps](#) are effective communication tools. FBA prepared a storymap about [Spatial Business Planning](#) as an example of how this powerful tool can be used with geospatial analyses for program or business planning efforts. However, it is not open source and can be difficult for some to access. For this reason, we have often switched to Jupyter Books for storymapping articles.
- **Geospatial analysis tools** such as QGIS, Google Earth Engine, R, and geospatial Python libraries, e.g., Geemap, Leafmap, are critical for planning forest conservation and businesses using spatial data and analysis. Looking for ways to integrate program design to guide geospatial analysis and the analysis to ratify or measure components created in the program design is a beneficial way to incorporate spatial data into planning.

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Proposals

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Iterative Design

The project and proposal development workflow is iterative, like program design, similar to the budget process described in the budget section. In general, we suggest the following workflow to develop projects or a proposal:

1. **Brief.** Create a 1-2 page brief with the following sections:
 - a. Executive summary. Three to five sentences max outlining the reason for the project, tasks, outcomes, timeline, and high-level budget.
 - b. Project goals. The project links to the programmatic theory of change and goals, followed by project goals/objectives.
 - c. Tasks. These could be listed under project objectives and also called interventions or activities.

Don't worry about fleshing out these sections in detail. When drafting the brief, focus on getting the ideas down. The details can be developed later.

Tip

Nest your goals, objectives, and tasks similar to an outline:

1. Goal 1
 - A. Objective 1
 - a. Task 1
 - B. Objective 2
 - b. Task 1
 - c. Task 2
2. Goal 2
 - A. Objective 1

1. **Budget.** Creating a back-of-the-envelope budget helps quickly estimate a project's basic costs—like staff time, equipment, materials, and key activities, e.g., site assessments, community outreach, training sessions, and report preparation—before investing time in a full proposal. It's a simple way to test feasibility and guide early planning decisions. Again, don't worry about details, just make some rough estimates. Napkins (or junk mail) and a pen at a restaurant are good tools of the trade! Capture your proposal ideas and get rolling.
2. **Scope.** Draft the proposal narrative or scope of work with a detailed description of tasks and activities by objective, using the brief from step one as a guide. At this point, it is worth adding a GANTT chart as a project timeline to show the project workflow and check if you've stacked too many activities at certain times or if your schedule is too aggressive.

Scope

Use the brief to flesh out a more detailed project scope and description. Don't be afraid to write and then rewrite various sections. Give each a rest for a few days, then return after working on other sections. Create a checklist from the proposal RFP solicitation package to ensure you don't forget anything. The [Forest Business Alliance](#) has a template checklist that may be useful.

Send your proposed ideas to colleagues for review or try to develop them across an interdisciplinary team or collaboratively if working with multiple partners.

Budget

Identifying and reviewing a proposal's requirements and other set project parameters early will help focus project design efforts and keep expectations in line with reality. One key parameter is the amount of money available for direct program implementation. Starting with a rough estimate of the required fieldwork will help you make realistic project scope and scale decisions.

Say you're developing a \$1 mn budget project, dividing it into broad categories using a budget calculator (Table 1). In this simplified example, salaries (35%) include benefits and contracted labor. Implementation (50%) is for all non-salary project costs, such as supplies, equipment, and travel. Indirect (15%) is for an organization's operating costs, e.g., rent, utilities, and fees. This immediately informs the project team how to allocate resources to the budget and project as they develop the proposal or project.

Table 1: Back-of-the-envelope budget calculator.

The calculator will give you rough amounts for each broad category. You could also break it down by your objectives and tasks from step 1 and the rough budget.

Warning

Critical: include inflationary increases in your budget, especially for salaries and material costs that will increase over the project lifetime (at least 4%/yr). If an application only allows one number per item, calculate costs in your budget spreadsheet over the number of years of the project.

Budget process

The project budget is the core engine of the project. Developing it with defensible numbers is critical to project success.

1. **Tasks.** Create draft objectives and tasks.
2. **Budget rough draft.** Draft a back-of-the-envelope budget by project category or deliverable, roughing out the approximate costs with rules that divide out salary implementation.
3. **Narrative.** From the draft tasks and budget, write a proposal narrative.
4. **Timeline.** Create a project GANTT chart with deliverables from the budget and narrative.
5. **Refined budget.** Return to the budget and refine numbers—test assumptions about estimates for each line item. Increase amounts as necessary. You may need to factor in inflation, salary raises, or increases in material purchase amounts that may fluctuate with changing markets. Adjust the total budget to

reflect what you think is a reasonable ask to the funder. This may require factoring some estimates down.

Tip

Creating budgets by scratch using a spreadsheet can be tedious, especially when you may need to submit multiple budgets in different formats. If you use Excel, create a table with your budget details. Select the rows/columns in the table, then Styles>Format as Table, and choose the table format colors you like. Here's an example:

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in your region



Then, with your cursor in one of the table cells, select Insert>PivotTable>OK. Click in the pivot table report field and, in this case, click and drag Category and Task to the Rows field, Year to the Columns field, and Total to the Σ Values field (with some number formatting and dragging Communications and Reporting to the bottom):

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in your region



To build out your budget, add more rows to the detail sheet, repeating activities by year. After adding new rows, always click Data>Refresh All to update the summary table(s).

Finalize

Put together your project design and share it. For both proposals and projects, copyedit your final version and then copyedit it again. Look at it in different formats and, ideally, let the proposal rest for a few days, then copyedit again. Printing out the key written components and editing with a red pen can also help catch additional mistakes or grammatical errors. This might seem like a paper waste, but it can be effective for editing.

Recommendations to Funders

Putting together proposals in response to foundations and government agency requests for proposals (RFPs) can often be an arduous or torturous process. It doesn't have to be this way, and some foundations and agencies are much better at it than others. For example, the California Wildlife Conservation Board

typically requires a site visit prior to an applicant putting together a proposal, then program officers review submitted proposals and provide feedback on scope and budget. If and when the proposal goes to the board, it's likely to be funded.

Funders vary so widely it's hard to make generalizations, but the best pay more attention to process and capacity building. If you get a chance to give feedback to funders during a proposal process, please offer them honest feedback. Here are our suggestions for funders that you're welcome to adapt:

1. **Streamline the initial application process.** One page: tasks, budget, qualifications. That's it! Details later! Don't make grantees answer 17 questions and develop a 50 page proposal for funding they may have a small chance of securing. That's not fair to them or reviewers.
2. **Consider participatory grant-making.** What if we flattened the decision-making for who gets funding and included grantees in that process. What if grantmaking by all funders was more transparent and less opaque? Maybe this would require more time up front by funders, but it could lead to stronger projects and improved connections to grantees.
3. **Rethink proposal forms.** Consider providing templates in word or pdf format that grantees can fill out then upload the saved docs to your grant portal instead of using online forms that require filling out each section before moving on to the next. No one should have to cut and paste their proposal to submit it online; that is so 1990s. Look for others ways to automate this process with grantees.
4. **Shorten RFPs.** RFPs in excess of 20 pages make it difficult for grantees to get to translate to proposals. List what you want, how much to apply for, and leave the legalese for funding recipients, e.g., don't include pages of appendices with contracts and required forms. The legal documentation can have a link online and providing other key documents, e.g., model budget or scope could be made available.
5. **Set realistic due dates.** Not January 2 or the week after Thanksgiving or even during the summer school holidays. Also don't release RFPs on December 23rd, that's verging on Ebenezer Scrooge behavior. These due dates aren't made up, we have experienced this for grants and heard from partners about the same.

The FBA can also review projects and proposals, with our staff writers' years of experience with proposal development and successful funding track records. However, we ask that you send them early, e.g., months before the due date or during the conceptual phase. Please get in touch with us if you're interested in a review.

Good luck!

Resources

The project design chapter is a very brief look at how to develop projects and proposals. You'll want to dive deeper to develop your project and proposal craft. Here are a few additional resources to get going:

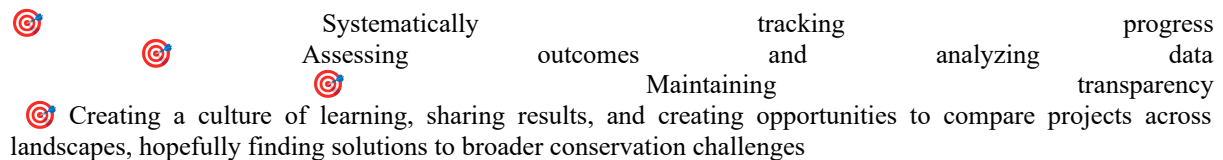
- [Forest Business Alliance](#). Templates upcoming workshops to prepare you for proposals and past workshop recordings on various aspects of program, project, and proposal development.
- [Grammarly](#). Paying for a better grammar/spell checker (than those available in Word or Docs) may help improve your writing.
- [Sierra Nevada Conservancy Grant Resources](#). SNC offers a grants newsletter and regular grant writing workshops. If you're in the Sierra, contact your local Area Representative to learn more.
- [North Coast Resource Partnership](#). NCRP offers technical assistance for proposals and project development and currently offers weekly office hours for real-time assistance. See the no-registration Zoom link on their What's New page to join.

Monitoring

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Background

Monitoring, Evaluation, Accountability, and Learning (MEAL) is crucial for ensuring the effectiveness and success of projects, as well as long-term organizational and business viability. MEAL enables organizations to make informed decisions and adapt strategies in real time by



This comprehensive approach helps identify and address issues early and fosters a culture of continuous improvement. Effective MEAL practices enhance accountability to stakeholders, ensure resources are used efficiently, and contribute to the overall sustainability and impact of projects. Despite the importance of MEAL, donors rarely provide adequate funding for it. Given this reality, you must find ways to implement MEAL with whatever resources—personnel, time, and budget—are available. This quick-step guide will get you started.

Definitions

-  **Monitoring** involves tracking project activities and progress.
-  **Evaluation** assesses the effectiveness and impact of the project.
-  **Accountability** ensures that the project is transparent and responsible to stakeholders.
-  **Learning** refers to using insights from monitoring and evaluation to improve future projects and practices.

Takeaways

Key points to monitoring, evaluation, accountability, and learning on a shoestring:

- **Big picture first.** There is a tendency to jump straight to metrics and indicators before developing a theory of change and programmatic goals and objectives. Monitoring plans should consider the big picture, so go through the process of developing the program strategy first. However, once created, feel free to change items, such as false assumptions or critical pathways that don't reflect reality.
- **Assessment & learning.** Link your assessment or theory of change to your monitoring and learning, e.g., Theory → Monitoring → Learning.
- **Time to learn.** Build learning into our organizational culture. Plan a retreat to analyze results and write about project learning that you publish in blogs, posts, and journals. Build learning into project closing and developing new projects.
- **Big investment/big rewards.** Building a successful measurement system takes time, staff capacity, and investment. However, the rewards are huge, and the cost dramatically decreases once the system is in place. Don't fall into the trap of processing, analyzing, and sharing data near the end of a project or phase.
- **Participatory research and measurement.** Partner with science-based organizations to increase your learning and expand capacity, or use citizen science to engage and connect the general public to your mission and collect data/demystify conservation science at the same time. Employing participatory approaches at all levels of evaluation also takes more time but makes for more meaningful, longer-lasting, and effective results in the long term.

Challenges

Monitoring could be the subject written about the most but implemented the least in the nonprofit world. Why is this? The challenges involved give an insight:

- Monitoring is hard to do and is hard to do right.
- Monitoring is boring.
- Your organization doesn't have staff capacity or applied scientists who can conduct monitoring.
- Project funding doesn't include monitoring, and it can be costly.
- It could demonstrate that your theory of change, project assumptions/interventions, or organizational mission is incorrect. That's a big elephant in the room.
- Your organization lacks a learning culture.

Solutions

How can these challenges be overcome without requiring massive institutional investment and resources? Let's start with the big picture of starting with the whole and work through other recommendations.

Start with the Whole

It is critical to work from the big picture to your metrics. Follow a process and be deliberate about creating your success measurement system.

- **Situation assessment.** Utilizing metrics without a measurement system or situation assessment is like building a house without a blueprint (see [Program Design](#)). Start with a situation assessment and understand how you, your staff, team, and partners understand the current contextual situation of the system you're working in and how your project interventions or business application will positively change that system. Do not just assume everyone gets what you're going to do. For example, if you ask the ten people you work with to define restoration, you might get ten different responses. Memorializing the situation, debating the relationships, and sharing them reach that common understanding.
- **Results chains.** Create results chains from your interventions to project outcomes. For more details on results chains, see the [Program Design](#) chapter and the [Open Standards](#).
- **Metrics.** Once results chains are established, add how you will measure intermediate and long-term outcomes.

Workshopping these steps, preferably with staff and partners, can get you to a cost-effective system sooner. Don't worry if it seems to take a while at the outset. Socializing across staff and partners can seem to delay the process, but it will ultimately create longer-term solutions.

Tip

Invite existing or potential funders to a planning workshop. There's no guarantee this will lead to funding, but you will certainly engage them, make your project more memorable, and tap into their deep knowledge regarding project success.

Utilize Existing Data Sources

Review all your existing data sources, as well as free data sources, before you invest in any new data collection efforts. Organizations often collect relevant data that could be used for M&E, but staff responsible for monitoring and evaluation must be made aware of it. For example, if you are seeking to measure return on investment (ROI), your finance team might already have information you can use to crunch ROI figures. Also, investigate open-source data, such as government and citizen scientist databases, before collecting more information. For example, a community forestry initiative seeking to gauge the fire resistance of a stand it manages could search available USGS datasets to find previous fires that may have impacted its site.

Employ Participatory Approaches

Engaging stakeholders in the M&E process enhances data validity and relevance and reduces data collection and analysis costs. Participatory approaches empower community members, beneficiaries, and partners to contribute their perspectives, insights, and local knowledge, enriching the M&E process. For example, a nonprofit organization conducting a program evaluation might organize focus group discussions, community meetings, or participatory workshops to gather feedback from program participants and community members, thereby reducing the reliance on expensive external consultants.

Participatory approaches may also employ apps to crowd-source data and engage key audiences in your mission.

Embrace Technology

Technology offers cost-effective and time-saving data collection, analysis, and reporting solutions. Many open-source and low-cost software, mobile applications, and digital platforms can streamline M&E data collection, facilitate data analysis, and enhance data visualization Russell (2024). For instance, a conservation organization might use a mobile data app such as Survey123, QField, or Kobo Toolbox to collect real-time forest health indicators, utilize GIS software such as ArcGIS Pro and QGIS to analyze, and use cloud-based platforms to store data, eliminating manual data entry and paper-based reporting. App-based data collection also allows multiple users to collect and input data.

ArcGIS Survey 123, from ESRI, is a good example of a mobile app for field data collection and mapping. It can be customized to collect various forest metrics such as tree species, diameter at breast height (DBH), and tree condition, and this information can be fed directly into ArcGIS software. It is easy to use and synchs effortlessly to your desktop or online GIS software. However, it does require a subscription, which can be expensive for some organizations.

Emerging low-cost, high-tech monitoring options

Traditional methods for monitoring forestry, ecology, and fuels can be expensive, time-consuming, and prone to errors when deployed at scale. Additionally, conventional methods can be slow to implement, limiting their usefulness in detecting new ecological changes, and they often do not provide data with enough precision or in a format that can be easily analyzed without extensive human systematization. Despite these limitations, many forest managers continue to use traditional methods because high-tech methods have historically been too costly to deploy. Fortunately, relevant technologies – including terrestrial laser scanning (TLS), unmanned aerial vehicles (UAVs), and light detection and ranging (LIDAR) systems – are rapidly becoming affordable options due to advancements in manufacturing processes, increased competition, and economies of scale.

USDA and USFWS researchers have successfully demonstrated using relatively low-cost (~25,000 USD), off-the-shelf TLS units like the Leica BLK360 for forest monitoring Pokswinski *et al.* (2021). Their methodology combines TLS calibrated using traditional, physically sampled forest conditions to simplify and accelerate the extraction of forestry, fuels, and ecological data from a single TLS point cloud. It streamlines data collection, reduces sampling bias, and generates easily processed data for improved modeling and decision-making.

Other researchers have developed a very low-cost system (~1,500 USD) that integrates DJI Livox MID40 laser scanner with an off-the-shelf UAV, and evaluated its capability in estimating both individual tree-level and plot-level forest inventory attributes Hu *et al.* (2020). They comprehensively compared their low-cost and four high-end UAV lidar systems equipped with high-end laser scanners to survey coniferous and broadleaved forest sites. The DJI Livox MID40 system performed comparably to the more expensive systems. Forest researchers and managers will struggle to resist as technology improves and becomes even cheaper.

Build Internal Capacity

Consider training existing staff to build internal M&E capacity, reducing your reliance on costly external consultants and service providers. By equipping staff with the necessary skills and knowledge, organizations can conduct M&E activities in-house, saving costs and ensuring sustainability. A small non-profit organization might provide M&E training workshops for its staff, empowering them to design surveys, analyze data, and generate reports independently, enhancing organizational efficiency and effectiveness.

Foster Collaborative Partnerships

Collaborating with other organizations, academic institutions, and government agencies can facilitate resource-sharing, knowledge exchange, and joint M&E initiatives. By pooling resources and expertise, organizations can achieve more comprehensive M&E outcomes at a lower cost. For example, a consortium of NGOs working on a regional conservation project might establish a joint M&E framework, share data, tools, and methodologies, and coordinate evaluation activities to avoid duplication and maximize efficiency.

Seek Creative Ways to Fund Monitoring

Many funders don't provide funding for monitoring, but they almost always require project reporting. Incorporating your measurement into reporting may help cover the costs. Educate the funders, too. For example, don't let them require reporting on 'acres and bucks' and explain why those metrics are okay, but don't tell the full picture about complex states such as forest health or resilience.

Going back to partnering with other organizations is critical here. For example, you could partner with a university on a research project. First, create a research agenda for your work to point researchers to what data and information you need (rather than them coming to you with pre-formed research). Academic researchers will want specific outcomes that might not connect with your data and outcomes needs, but they could help develop technology or collect some data you need.

Car Analogies

Originally, we framed organizational monitoring approaches as the extremes of owning a Yugo vs. a Cadillac. This is a poor analogy since Yugos were cheap, broke down frequently, and some consider them the worst cars ever. On the opposite end of the spectrum, Cadillacs are expensive and beyond the purchasing power of many individuals.

When I was at university, starting my senior or fourth year, I owned a simple red Honda Civic hatchback. It was plain but incredibly dependable and gas efficient. I never had a single-engine problem with that car, and it always started, even during the dark, cold months of winter when some autos from the 1970s and 80s rebelled and would not turn over in frigid northern US temperatures.



On the other hand, you may need something equally dependable, affordable, and expansive, such as the workhorse minivan. A Civic will get you reliably where you need to go, whereas a min-van will do that and allow you to bring along your kids, their friends, and all their stuff. I rented a Kia minivan once when the rental agency at the San Diego airport ran out of cars. That thing was a rocket. I had it going 90 mph several times (don't tell CHP) while driving through the Cleveland National Forest. Vroom vroom!

What is the absolute bare-bones, dependable monitoring system that could be implemented (Civic) vs. a more expansive version (Minivan)? We suggest a couple of examples in the pulldowns below. Both examples are hypothetical stand-thinning projects with prescribed fire applied periodically after thinning for maintenance. The project assumption is that thinning will lead to increased forest health and reduced high-intensity wildfires. Sometimes, you need a dependable Honda Civic:

Honda Civic Monitoring Methods Checklist

Intervention is stand thinning to reduce wildfire risk followed by regular prescribed fire treatments.

1. Stand inventory by species and dbh every ten years using Kobo Toolbox and QGIS
 2. Photo witness posts at reference and treatment sites
 3. Total acres treated, total acres burned
 4. Total dollars spent

And sometimes you need a Minivan:



Minivan Monitoring Methods Checklist

Intervention is stand thinning to reduce wildfire risk followed by regular prescribed fire treatments.

1. Stand inventory by species and dbh every ten years using Survey123 and ArcGIS Pro
 2. Photo witness posts at reference and treatment sites
 3. Total acres treated, total acres burned
 4. Total \$s spent
 5. [Audiomoth](#) IoT set up in stand to monitor birds and bats
 6. Pollinator survey once/yr to morpho spp. to measure the impact of thinning and fire on invertebrates
 7. Understory vegetation survey every five years

Resources

- **[Recipes for Conservation](#)**. A summary of how to develop the conservation standards within or among organizations. The recipes have some valuable sections for businesses but are nonprofit and agency-focused.
- **[Pause & Reflect Toolkit](#)**. Pause and reflect is an organizational stop and step back to reflect on what you're doing and why as well as an evaluation of how it's going. This toolkit provides project leaders with a centralized collection of resources, enabling them to find everything they need to lead and support a pause and reflect process.
- **Measures of Success**. Practical guide (in English and Spanish) for designing, planning, and implementing conservation projects. The Spanish version can be found as a free download, and the English version is available for purchase from Island Press Margoluis & Salafsky (1998).
- The **Open Standards for the Practice of Conservation**, developed by the Conservation Measures Partnership is a definitive resource for program and project design (CMP, 2025).

References

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eCivis

Keri Rinne, Kelly Kahling

Background

CAL FIRE uses an online system called eCivis to receive applications and administer grants. You will use this system to apply for a CAL FIRE grant and manage and report on that grant if your application is successful.

The following information covers two stages at which you will use eCivis: The proposal stage, during which you will create a profile and submit all proposal materials, and the post-award stage, during which you will manage your project and use eCivis to submit Activity and Financial Reports.

It is important to note that this guide is meant to be used in conjunction with the Business & Workforce Development resources provided on CAL FIRE's Wood Products and Bioenergy [website](#) and eCivis-specific materials. In particular, the Grants Portal Application Submittal User [Guide](#) and the eCivis Grants Portal Applicant Troubleshooting [Guide](#).

Proposal Stage

Access the eCivis portal for the Business & Workforce Development Grant Program through the link in the CAL FIRE guidelines for the solicitation (Figure 1). Note that this link is updated for each solicitation, so we are not including it here. For the Wood Products and Bioenergy Program, the solicitation guidelines are posted on the right side of the page under 'RECENT UPDATES'. It is important to create your account using the link in the grant guidelines. This will ensure you access the correct website and create an application in the correct solicitation window. Your account will be used for your application and to administer your grant, should your application be awarded funding.

APPLICATION PROCESS

These processes are applicable to both business and workforce development proposals unless otherwise specified.

Applicants to Wood Products grants apply through the Grants Portal in eCivis (linked [here](#)). After you create an account, use the "Portal Login" and complete the application, including uploading all required documentation using the Grants Portal. Incomplete applications may be disqualified. Guidance for setting up a new Grants Portal account via eCivis can be found within the *eCivis Grants Portal - Application Submittal User Guide* and also accessible on the [CAL FIRE Grants Webpage](#). No hard copy or emailed applications will be accepted. The eCivis Grants Portal may close occasionally when CAL FIRE needs to update available funding information, update the grant guidelines, or revise the application format.

The information required in the application form will differ depending on the type of project and consists of several sections:

Figure 1: Application process.

Clicking the link in the first sentence of Figure 1 takes you to a landing page with summary information about the solicitation (Figure 2). Before beginning your application in eCivis, download all the required forms and templates you will fill out and upload as part of your complete application (e.g., Work Plan and Environmental Compliance Worksheet, State of California Payee Data Record, Non-Discrimination Compliance Statement). These are provided on the Grantee Resources section of the Program website and at the Files tab on the far right of Figure 2.



California Department of Forestry and Fire Protection (CAL FIRE)

Business and Workforce Development 2025 Q1

☒ Apply

Overview

Eligibility

Financial

Contact

Files

ID:

N/A

Title:

Business and Workforce Development 2025 Q1

Application Start Date:

02/20/2025

Application End Date:

04/26/2025

CFDA/ALN:

N/A

Reference URL:

Summary:

Funding for the 2025 Q1 Business and Workforce Development Grant solicitation is part of the California Air Resources Board (CARB) California Climate Investments (CCI). CCI puts billions of dollars to work reducing greenhouse gas (GHG) emissions, strengthening the economy, improving public health and the environment, and providing meaningful benefits to the most disadvantaged communities, low-income communities, and low-income households.

* eCivis Inc. is not responsible or liable for user-generated content.

Figure 2: Initial solicitation landing page.

Click the Apply checkbox below the CAL FIRE logo, leading you to a login page (Figure 3). Click Create an Account if you do not already have one, or log in if you do. Use the portal login.

Welcome to the Portal



This service provided by eCivis makes it possible for interested parties to save, collaborate, and apply for solicitations. Login, or create a free account to start.

(Minimum 8 chars, alphanumeric with symbol(s))

Portal Login

Grants Network® Login

Forgot Password?

Don't have an account?

Create an account

eCivis Grants Network user?

Use your existing login above and the Grants Network® Login button.

Figure 3: eCivis system log-in page.

This will lead you to the submission portal, where you can start a new application or continue working on a previously saved application (Figure 4).



California Department of Forestry and Fire Protection (CAL FIRE)

Wood Products and Bioenergy

Business and Workforce Development 2025 Q1

My Applications

Create New Application

Show 10 entries

Search:

Grant Application	Create Date	Status	Actions
No applications have been saved or submitted			

Showing 0 to 0 of 0 entries

Previous

Next

Reload

Figure 4: The eCivis submission portal.

Created with  Curvenote

1. Don't forget to access and use the templates at the [Forest Business Alliance website](#). FBA team members are available to assist you with this. Contact us at forestbusinessalliance@gmail.com if you have questions or need further assistance.
2. The Forest Business Alliance's guidebook [Chapter](#) on project and proposal development would be helpful at this stage for organizing your proposal narrative and budget development workflow. Before you start a proposal, review the chapter to establish how your proposed project fits into your business or strategic plan.

Follow the process in the Grants Portal Application Submittal User [Guide](#), which will take you step-by-step through creating an account and completing an application.

eCivis Budget Tips

- Use the budget template on the FBA template [page](#) to develop your project budget. As you complete the Detailed Budget worksheet (step 3 in the template), the template will automatically populate a worksheet (Output B. eCivis Upload) that can be used to complete the eCivis budget form. FBA created the template to make it as easy as possible to go from calculating a budget in Excel to filling out the eCivis form.
- Please refer to the project/proposal chapter of the FBA guide for tips on developing your budget and the relationship between budget and scope development.
- If you are applying on behalf of a non-profit or Tribe, you may be able to charge indirect costs. Other types of applicants, e.g., businesses, cannot charge indirect costs. If you have questions about indirect costs or how to include them in your grant budget, refer to the grant solicitation guidelines and the eCivis template.
- Project budgets in eCivis are shown in categories such as Salaries & Wages, Contractual, Travel & Per diem, Equipment, Other Costs, Indirect Cost. To enter budget items into each category, click on the category name (e.g., Salaries & Wages), which will bring up additional boxes to fill in. Line items automatically calculate the unit and unit cost the first time you enter them. However, the line item will not automatically update if you change one of these amounts. You will need to recalculate the Costs manually and, if applicable, the Cost Share.
- Both extended and cost are calculated by multiplying units by unit cost and are not distinguished in eCivis.
- Budget Narrative
 - The budget narrative is entered at the bottom of the budget page on eCivis.
 - It has a limit of 8,000 characters, which includes spaces.
 - You should create your narrative in Word and then cut and paste it into eCivis so that you have a record of the narrative you submitted. It cannot be downloaded from eCivis once submitted.
 - For guidance related to budget narrative content, please see the Forest Business Alliance Budget [template](#).
 - Be specific about line items and your justification for including each in the budget for your narrative.

Post-award Stage

The eCivis Grants Portal CAL FIRE Grantee User Guide provides an overview of all stages of post-award grant management through eCivis. If you have been awarded a Business and Workforce Development Grant, this document provides the basic steps you need to take, from accepting the grant award to adding team members to your online portal who will assist with reporting or invoicing and how to submit financial and activity reports.

Reporting

CAL FIRE requires a minimum of quarterly reporting; however, you can invoice and report as often as monthly. Below is a description of the reporting process, followed by a description of the financial reporting.

Invoicing and reporting involve a three-step process:

1. Complete the online Progress Report Form. CAL FIRE will provide a template and guidance on this document as part of your onboarding process if you are awarded a grant.
2. Complete the Activity Report in eCivis (described below).

3. Complete the Financial Report in eCivis (described below).

Activity Reports

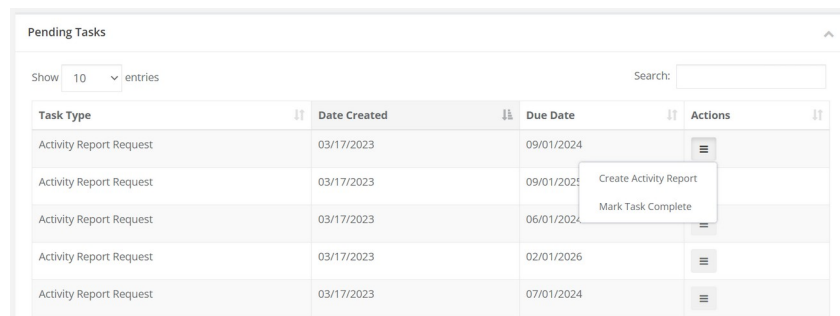
Click on “My Awards” on the left side of the screen. A list of your awarded grants will appear. Under the “Actions” tab, you’ll see a button with three lines. Click it, and then click “Manage Award.” This will take you to your Award Dashboard.

Note

Note: The Activity Report must be completed in addition to the Progress Report Form, which you will submit online via Microsoft Forms.

There are two different ways to create and submit Activity Reports:

1. Locate the Activity Report due under Pending Tasks and click the button with three lines, then click “Create Activity Report.” (Figure 5). This will take you to the Activity Report page, where you will enter the reporting period, narrative, and metrics and upload any deliverables for the reporting period.



Task Type	Date Created	Due Date	Actions
Activity Report Request	03/17/2023	09/01/2024	⋮ Create Activity Report Mark Task Complete
Activity Report Request	03/17/2023	09/01/2024	⋮
Activity Report Request	03/17/2023	06/01/2024	⋮
Activity Report Request	03/17/2023	02/01/2026	⋮
Activity Report Request	03/17/2023	07/01/2024	⋮

Figure 5: Option 1 for creating and submitting an Activity Report is to access it under Pending Tasks on the Award Dashboard page on eCivis.

OR

1. Look for the “Submit Activity Report” button (in green) at the top of the Award Dashboard (Figure 6). Either enter the due date of the pending Activity Report or click “Continue with New Report.” Either option will take you to the Activity Report page. Enter the Reporting Period, Report Narrative, and Activity Report Metrics. Leave the metrics section blank if you have not completed any goals or objectives during the reporting period. Upload any deliverables at the bottom of the page. If you are ready to submit, click the red “Submit Report” button. You cannot undo submitting the report, so only click this button if the report is final. If you want to save the report as a draft and return to it, click the blue “Save & Close” button. If you have saved a draft, you can access it again in the “Award Activities” section.

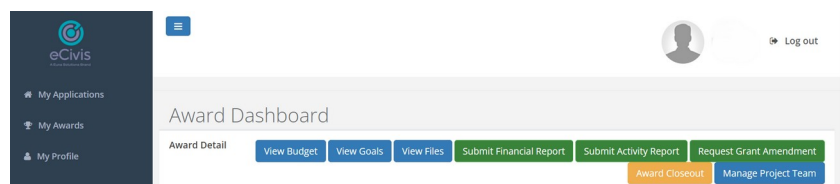


Figure 6: Option 2 for creating and submitting an Activity Report is to click on the green button that says “Submit Activity Report.”

Financial Reports

A Progress Report and Activity Report must accompany financial reports. At the outset of your project, it's important to consider your needs regarding invoicing, as this will impact the frequency of all of your reporting. For example, if you submit monthly invoices, you must also submit Progress Reports and Activity Reports.

A helpful first step in financial report preparation is to review your current project budget totals, which can be seen by clicking on the blue “View Budget” button from the Award Dashboard page (Figure 6). If you want an Excel version of the information, click the small Excel icon at the top of the page next to “Actions.”

When submitting an invoice to CAL FIRE, you must prepare and sign the CAL FIRE Invoice Form (provided upon grant award; an example can be found here). You will save and submit this as a standalone one-page PDF file. Additionally, you will be required to prepare a separate PDF file packet for each Budget Item listed on your CAL FIRE Invoice Form for which you are requesting payment.

These PDF file packets contain invoicing backup documentation for your expenditure reimbursements, plus any receipts or pertinent supporting information that should accompany it. CAL FIRE requests that the file names correspond to the Budget Item category. It's also helpful to include your invoice number, such as “CONTRACTUAL – INV-07” or “TRAVEL – PMT-01.” For example, your PDF file “CONTRACTUAL – INV-07” invoice packet will contain all invoices and reimbursement requests for the invoicing period that fall under the Contractual category of your project budget. In the Financial Report Files section, these file packets will be uploaded as individual sets of documents to eCivis. Note that you do not need to submit supporting documentation when reporting your matching contributions; however, you should keep it for your records in the event of an audit.

When you are ready to submit your Financial Report, click on the green “Submit Financial Report” button on the Award Dashboard, and then click “Continue with New Report” when prompted.

Your Financial Overview totals are displayed in the top section (Figure 7).

Figure 7: Financial overview. Note that amounts have been redacted for privacy reasons.

Below that, enter the reporting period for your reimbursement. This Reporting Period should match the Invoice Period you entered on your CAL FIRE Invoice Form. The Invoice Period must include all dates mentioned in all parts of your invoice. For example, if a person submits reimbursement for a receipt dated January 19, the Invoice Period must include January 19. It is reasonable to expect these invoicing periods to overlap from invoice to invoice.

The items below the Financial Report Details heading describe the different budget categories of your award (Figure 8). These items correspond to the Budget Items in your Project Budget and your CAL FIRE Invoice Form. In the Financial Report Details section, for each Category/Budget Item listed, you will enter all reimbursements you request during the Invoicing Period.

- Your budget category file packets may likely contain multiple invoice activity requests. Each activity request should be entered on a separate line under the appropriate category. For multiple invoice activities, click the plus sign (+) to the right of the Description cell to add a row. If extra rows were added but not needed, these rows should be deleted by clicking on the minus sign (-).
- Pages 32-33 of the eCivis Grants Portal CAL FIRE Grantee User Guide explains how to complete the GL Code/Activity and Description cells.
- The bottom line, Report Total, will automatically tally your entered figures. This total should match the total on your CAL FIRE Invoice Form.

Category	Spend	Match	Award Approved	Match Approved	Award Remaining	Match Remaining	Total Remaining
1. Salaries & Wages	\$ 0.00	\$ 0.00	\$ 48,518.00	\$ 0.00	\$ 12,497.10	\$ 0.00	\$ 12,497.10
GL Code / Activity	\$ 0.00	\$ 0.00	Description				
2. Employee Benefits	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00
3. Contractual	\$ 0.00	\$ 0.00	\$ 934,176.00	\$ 0.00	\$ 610,863.93	\$ 0.00	\$ 610,863.93
GL Code / Activity	\$ 0.00	\$ 0.00	Description				
4. Travel & Per Diem	\$ 0.00	\$ 0.00	\$ 5,000.00	\$ 0.00	\$ 4,607.33	\$ 0.00	\$ 4,607.33
GL Code / Activity	\$ 0.00	\$ 0.00	Description				
5. Supplies	\$ 0.00	\$ 0.00	\$ 12,000.00	\$ 0.00	\$ 12,000.00	\$ 0.00	\$ 12,000.00
GL Code / Activity	\$ 0.00	\$ 0.00	Description				
6. Equipment	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00
7. Other Costs	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00
8. Indirect Cost	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00	\$ 0.00
Report Total	\$ 0.00	\$ 0.00	\$ 999,694.00	\$ 0.00	\$ 639,968.36	\$ 0.00	\$ 639,968.36
	0.00 %	0.00 %		0.00 %	0.00 %	0.00 %	0.00 %

Figure 8: Financial report details

For reimbursement requests, the box next to “Reimbursement Request” must be checked for the submittal to go through. The Financial Report Narrative is marked as mandatory, but it is unnecessary to enter extensive text here if there is nothing to report (Figure 9). A short sentence, such as “Activity Report submitted separately,” will suffice.

The screenshot displays the eCivis portal interface for submitting a reimbursement request. At the top, there's a header with a URL and a table of values (all 0.00). Below this is a 'Reimbursement Request' checkbox. The main section is 'Financial Report Narrative', which includes a text area for the narrative and a character count (8000 characters remaining). Below the narrative is the 'Financial Report Files' section, which has an 'Upload File' button and a table for uploaded files. The table currently shows 'No files are available for download'. At the bottom right, there are buttons for 'Save & Close', 'Submit Report', and 'Cancel'.

Financial Report Narrative *

Please develop your narrative below including key metrics, spending details, and other pertinent details.

8000 characters remaining

Financial Report Files

Please upload any files necessary for your financial report. This may include financial transactions, receipts, program income, etc...

Upload File

Show 10 entries

File Name	File Size	Actions
No files are available for download		

Showing 0 to 0 of 0 entries

Previous Next

Save & Close Submit Report Cancel

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Figure 9: Budget narrative.

The bottom section, “Financial Report Files,” is where you will upload your invoice file packets along with the CAL FIRE Invoice Form by clicking the green Upload File button and selecting the appropriate files from your computer. Once all files are successfully uploaded and all data correctly entered, click the red Submit Report button to submit your reimbursement request to CAL FIRE.

If you have any questions, please don’t hesitate to get in touch at forestbusinessalliance@gmail.com.

Compliance Pathways

Vance Russell, Andrea Williams, Kerri Rinne

Background

Compliance and documentation with the California Environmental Quality Act (CEQA) and the National Environmental Protection Act (NEPA) are required for all planned projects. Projects on federal lands require NEPA; everything else requires CEQA. However, most California state-funded projects may require CEQA regardless of project location. The CEQA Handbook for Bioenergy and Wood Products Businesses is a useful reference guide for project developers, local government, and business owners (Darlington & Moose, 2025)

CEQA

A general flowchart showing the decision-making process for the California Environmental Quality Act (CEQA) for any project on nonfederal land in California (Figure 1). This flowchart is a general CEQA process for any project requiring CEQA compliance. With forest health and wood products projects, please use the diagram as an overall environmental frame of reference rather than the exact process to obtain compliance for your project.

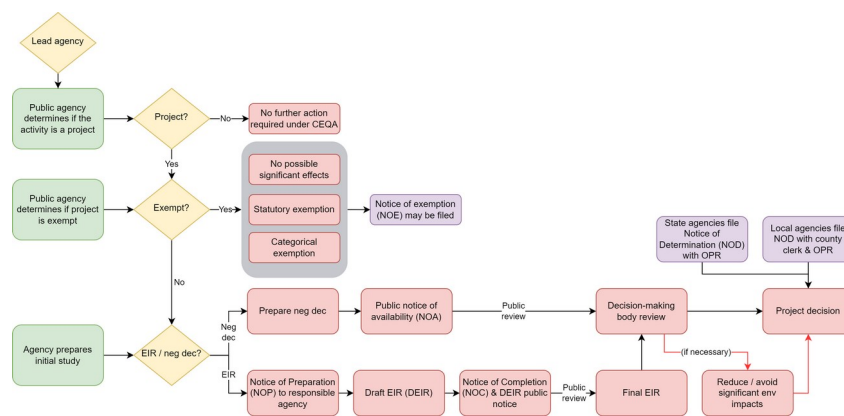


Figure 1: General CEQA flowchart (adapted from Rivkin *et al.* (2012)).

The types of CEQA documents are many but generally encompass the following broad categories listed from low risk/short completion time to high risk/long completion time:

1. **Notice of Exemption (NOE)** = The project is Categorical Exempt with no exceptions or has no possibility of impacting the environment.
2. **Initial Study** = Identifies project env impacts.
3. **Negative Declarations (neg dec/MND)** = The project will not have significant environmental impacts (neg dec) or has been redesigned to eliminate any significant impacts (Mitigated Negative Declaration, MND).
4. **Environmental Impact Report (EIR)** = Prepared when significant environmental impacts are identified.

CalVTP

[The California Vegetation Treatment Program](#) includes the use of prescribed burning, mechanical treatments, manual treatments (including reforestation), herbicide application, and prescribed herbivory as tools to reduce hazardous vegetation around communities in the Wildland-Urban Interface (WUI), to construct fuel breaks (shaded or non-shaded), and to promote ecological restoration. A general VTP process flowchart is shown in Figure 2.

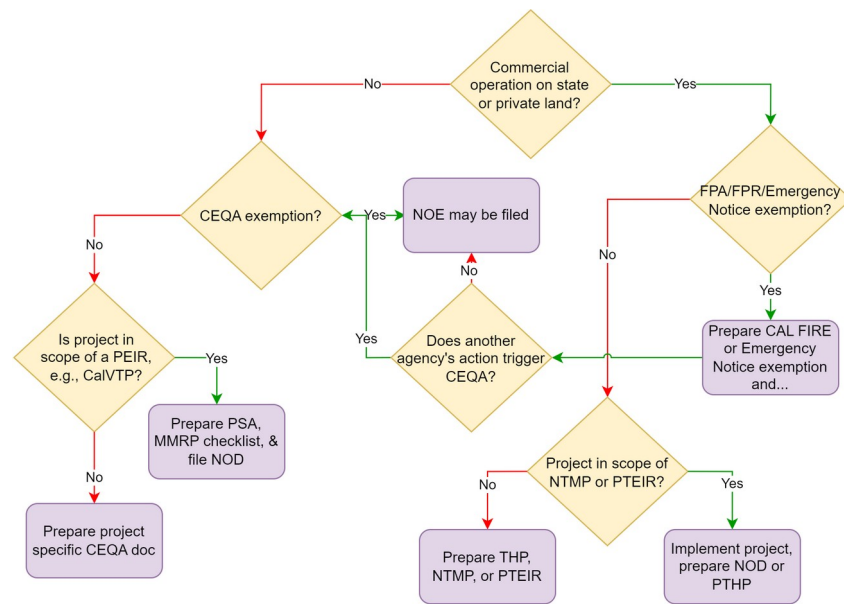


Figure 2: The CalVTP flowchart was adapted from ‘Environmental planning strategies to reach the million-acre goal’ (Ascent Environmental). See the glossary at the Chapter’s end for a key to acronyms.

Important timber/commercial definitions from California Code PRC 4527 include:

- (a)(1) “Timber operations” means the cutting or removal, or both, of timber or other solid wood forest products, including Christmas trees, from timberlands for commercial purposes, together with all the incidental work...”
- (2) “Commercial purpose” includes (A) the cutting or removal of trees that are processed into logs, lumber, or other wood products and offered for sale, barter, exchange or trade...”

Section IV - Cumulative Impacts Assessment of a Timber Harvest Plan (THP) - complies with CEQA. Once the THP is approved it is the functional equivalent of an EIR. The same environmental compliance applies to Nonindustrial Timber Management Plans (NTMPs).

Examples

A diagram showing compliance pathways for wood products and forest health projects (Figure 3).

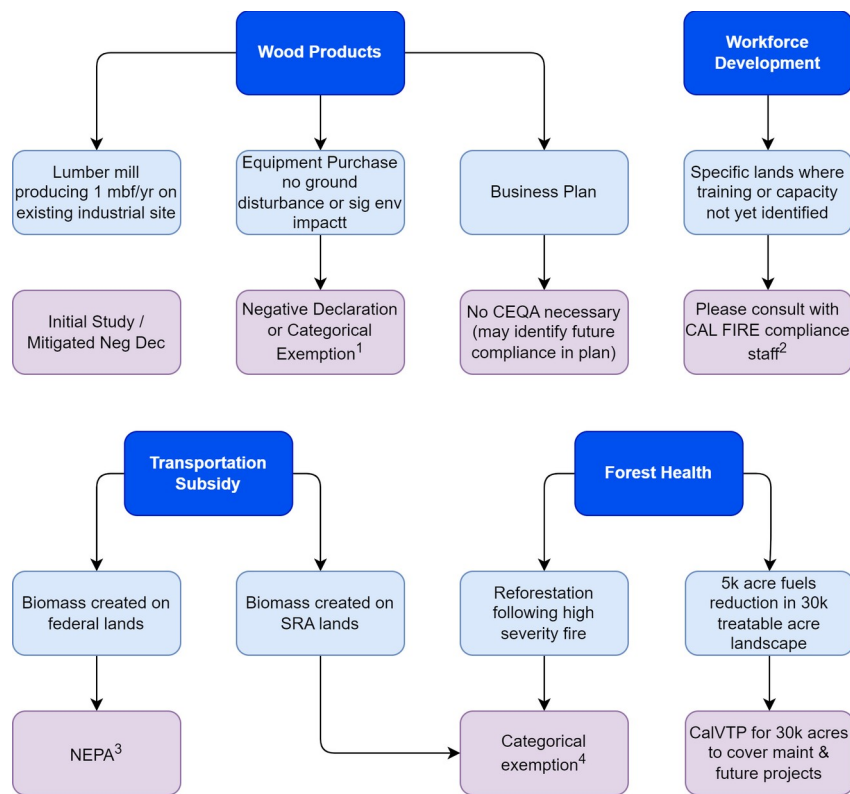


Figure 3: Example projects and CEQA/NEPA compliance. 1: Often added to an existing project where CEQA has been completed. In these cases, project proponents can use existing CEQA analysis and determine no new significant impacts via an addendum to the existing document; 2: Before the training, those lands must have completed CEQA and provide a copy of the analyses to CAL FIRE; 3: Projects submitting NEPA documents that created the biomass would be statutorily exempt from CEQA (AB211 - PRC 4799.05(d)(1)); 4: Projects could utilize the CEQA documents that created the biomass. In some cases, the transport is included in the facility analysis, so the facility impacts CEQA documents can be used if they include the biomass transport to the facility. See CEQA Statutory Exemption 21080(b)(3) and CEQA Guidelines Section 15269(a) for reforestation and emergency exemptions.

Glossary

- EIS/EIR - Environmental Impact Statement/Report
- FPA/FPR - Forest Practice Act/Rules
- IS - Initial Study
- MMRP - Mitigation Monitoring & Reporting Program
- NOD - Notice of Determination
- NOE - Notice of Exemption
- NTMP - Nonindustrial Timber Management Plan
- FEIR - Final Environmental Impact Report
- PEIR - Program Environmental Impact Report, PSA - Project-Specific Analysis
- PTEIR - Program Timberland EIR
- PTHP - Program Timberland Harvest Plan, THP - Timber Harvest Plan

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Resilience Gap

Veenu King, Michael Kent, Vance Russell

Wildfire

California's diverse landscapes encompass forests, grasslands, chaparral, and urban-wildland interfaces, each playing a critical role in the state's ecological and economic vitality. Its 33 million acres of forests span the Coast, Cascade, and Sierra Nevada ranges. Ownership is mixed across public and private, with 60% of forestland publicly owned—mostly National Forests followed by State and BLM lands USDA (2024). The remaining forest lands are split between private industrial and private non-industrial. The bulk of California forests are made up of mixed conifers—dominated by douglas-fir, giant sequoia, Jeffrey pine, pinyon, and ponderosa pine—followed by western oak woodlands CFF (2024).

These forests are fire-adapted. While wildfires are a natural element of California's forests, human activities, land use changes, fire suppression practices, and climate change have intensified the frequency and severity of wildfires (Figure 1).

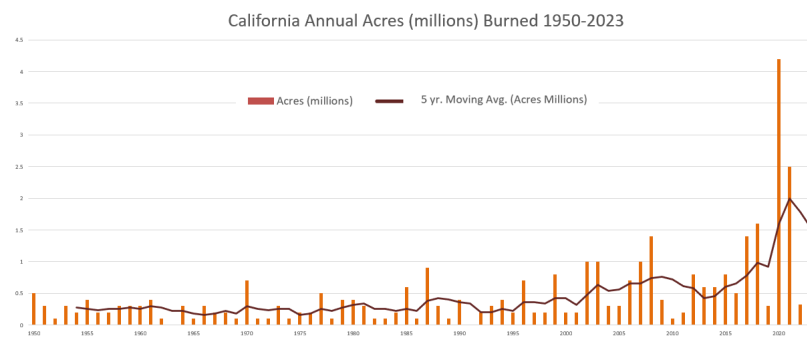


Figure 1: Annual acres burned statewide. Source: California Environmental Protection Agency's Office of Environmental Health Hazard Assessment OEHHA (2024).

In 2023, California experienced over 4,500 wildfires that burned through 116,399 acres. Midway through the 2024 wildfire season, the state has already experienced 5,383 wildfires burning through 827,880 acres FIRE (2024). This trend has shown no signs of slowing, with experts warning that the region has entered an era of megafires OEHHA (2024).

Disparities

California's megafires significantly destroy homes, infrastructure, and natural habitats, leading to loss of life, displacement, and billions of dollars in damages. They worsen air pollution, degrade water quality, and release large amounts of carbon dioxide, exacerbating climate change.

These impacts disproportionately affect low-income and Indigenous communities. Studies have examined the relationship between fire hazard ¹, fire experience, and income in California communities, showing that communities experiencing higher fire frequency have lower incomes (Figure 2). This result could be due to the lack of investment and resources in those areas or the repeated fires that have changed the communities over time.

¹Fire hazard is fire damage (property damage) whereas fire experience is frequency of fires. Median income in communities that have had 6 or more fires since 1990 is \$66,128 whereas those with 1-2 fires have a median income of \$100,063 (Hino and Field).

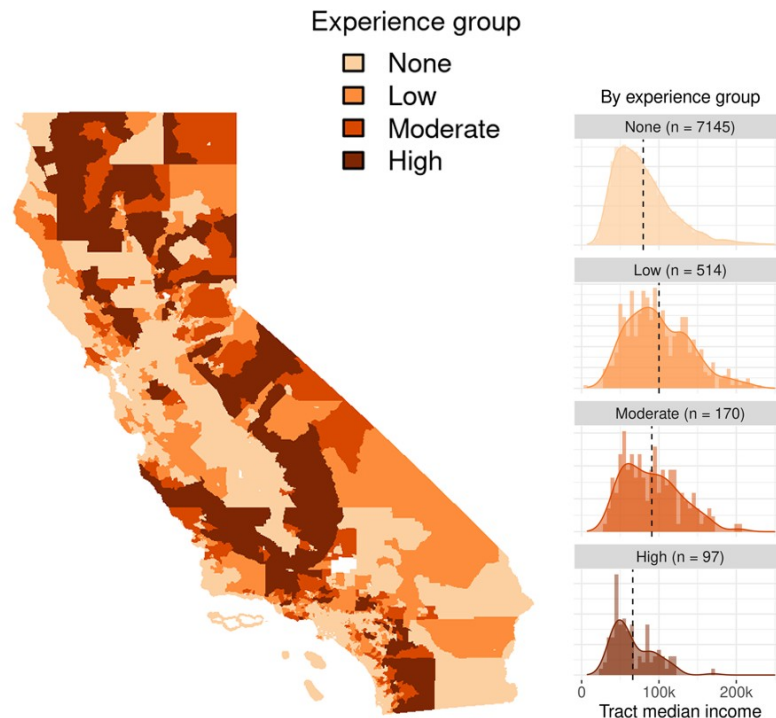


Figure 2: Fire experience by census tracts and corresponding median income by experience group. Source: Hino, Field: “Fire frequency and vulnerability in California.”

Additionally, this study’s analysis on fire suppression has found that communities with high property values stop the spread of wildfires, thus reducing their deleterious impact Hino & Field (2023). California’s low-income communities suffer repeated fires and subsequent health impacts from persistent air pollution as well as higher daily cardiovascular hospitalization Chen *et al.* (2024) Pohl (2024) Secaira (2023).

Treatments

California has identified more than 20 million acres that need fuel treatment across various landscape types to reduce hazardous vegetation. For this analysis, treatment needs are narrowed to private forest lands and privately owned forested areas with 20% or more canopy cover. Treatment for forests includes prescribed burning, selective harvesting and thinning, herbicide applications, and constructing fuel breaks. These interventions reduce fuels that generate catastrophic fire conditions and slow-burning fires, provide buffer zones to limit crown fires and return forest lands to a state that can support low-intensity fires. In collaboration with many federal, state, local, and non-government entities, the Forest Management Task Force published California’s Wildfire and Forest Resilience Action Plan, which aims to treat one million acres of state and federal land annually by 2025 WFRTF (2021).

Methods

This research focuses on the spatial overlap of wildfire treatment needs within private forests in low-income regions. QGIS was used to create spatial overlaps of different maps, including land ownership, poverty, treatment, and forest mapping datasets. Python was used for statistical analysis. Previous studies have examined the relationship between populations and past wildfires, but few have analyzed the forward-looking treatment needs at a census tract level Hatfield (2001) Shahir Masri *et al.* (2021).

Definitions

- **Forest and Wildfire Fuel Treatment:** Treatment includes prescribed burning, mechanical treatment, manual treatments, herbicide application, and prescribed herbivory to reduce hazardous vegetation, constructing fuel breaks, and ecological restoration Environmental (2024).

- Poverty Rate: California Environmental Protection Agency's Office of Environmental Health Hazard Assessment defines this as the percent of the population living below two times the federal poverty level from a 5-year estimate for 2015-2019 PPIC (2024).
- Poverty Population Threshold: High Poverty 20%; Extreme Poverty 40%. The USDA defines areas of "high poverty" as those with a poverty rate of 20%+ and "extreme poverty" as those with a poverty rate of 40%+ USDA (2023).

Datasets

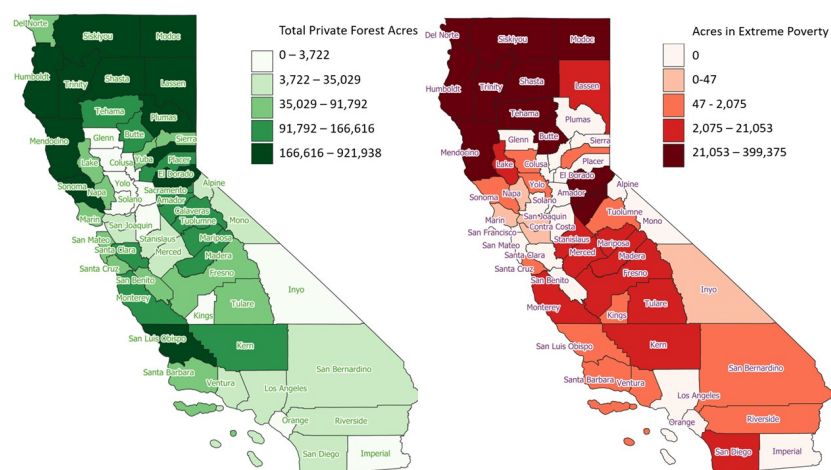
The following datasets were used for the study:

- Land Ownership Dataset: CAL FIRE's California Land Ownership dataset and the California Census Tract Dataset identify public lands.
- Forest Dataset: the 2021 National Land Cover Database USGS (2018). It is critical to note that the National Land Cover Database has identified three types of forests: Deciduous forests, evergreen forests, and mixed forests. To be labeled a forest, these regions' trees must comprise more than 20% of the total vegetation cover. This definition of a forest differs from that of USDA's Forest Inventory Analysis (FIA), which defines a forest as an area with at least 10% canopy cover and is for forestry use. As a result, the total acres of forest in California from FIA and NLCD are different, 19 million acres and 31 million acres, respectively. Additionally, in the NLCD dataset, there is no distinction between industrial private and non-industrial private forest lands.
- Forest and Wildfire Fuel Treatment Dataset: CAL Vegetation Treatment Program Environmental Impact Report and dataset California Vegetation Treatable Landscape
- Acres Treated Dataset: California Wildfire & Forest Resilience Task Force's Beta Interagency Treatment Dashboard. Most of the information captured in this dashboard related to forest treatment activities was used to identify the number of forest acres treated. There were 93,000 acres of targeted grazing (of the total 751,467 acres treated); however, it was unclear if it was forest-related and was included nonetheless.

Key Findings

This research has identified several critical insights that underscore the complex relationship between wildfire treatment needs and economic vulnerability in California. It has also provided a way to identify high-impact opportunity areas where financial assistance might have a greater socio-economic and environmental impact. Key findings include

- **More than 33% of the state's wildfire treatment needs are on private forest land.** Of the roughly 20 million acres of treatable lands across landscape types, 18 million acres are private lands; of that, 6.7 million acres are private forests (Figure 3).
- Of the 6.7 million private forest treatable acres, **4 million acres are concentrated in High Poverty Census Tracts.** That is, 60% of total private treatment needs are concentrated in areas where the poverty rate exceeds 20%—2x the national average.
- Of the 6.7 million private forest treatable acres, **over 1.6 million acres are concentrated in Extreme Poverty Census Tracts** (Figure 3). About 25% of total private treatment needs are located in areas where the poverty rate exceeds 40%—4x the national average.
- **Three of 58 counties make up half of treatable forests in Extreme Poverty Zones.** Mendocino, Humboldt, and Tehama have nearly 822,000 acres of private forest in Extreme Poverty lands requiring fuel reduction.
- **Current treatment levels are falling short.** In 2022, 86,000 acres were treated on private lands. If California continues at this rate, addressing the total private forest need will take more than 75 years.



Top 5 counties with the greatest total acres of private treatable lands in extreme poverty (poverty rate > 40), their acres in high poverty (poverty rate > 20), and their acres treated in 2022 (Table 1).

Table 1: Top 5 counties with the greatest total acres of private treatable lands in extreme poverty (poverty rate > 40%), their acres in high poverty (poverty rate > 20%), and their acres treated in 2022.²

Next Steps

The challenges California's diverse ecosystems face are compounded by the increasing frequency and intensity of wildfires driven by climate change. Low-income communities have been disproportionately affected by wildfires, experiencing higher health and economic burdens. This research confirms that low-income and underserved communities comprise most of California's fuel treatment needs but treat landscapes at rates that would take decades, if not centuries, to complete.

California can better safeguard its natural landscapes and most vulnerable populations by aligning wildfire treatment strategies with socio-economic resilience measures. Future studies and policy developments should prioritize these high-risk, high-need areas to foster a more equitable and effective approach to wildfire management and climate adaptation. Future research can delve deeper into granular information on the different treatment types across geographies, gather parcel-level data, and explore permitting, planning, and aggregation issues to conduct larger-scale fuel treatments in the state.

²Utilized data from the Wildfire and Forest Resilience Task Force’s Beta Treatment Tracker [Dashboard](#) to determine acres in treatment and percent of those acres that are privately owned WFRTF (2024).

Appendix

Table 2. Treatable privately owned acres in high and extreme poverty, total activity acres, and total activity acres on private land. ³

[illegible]³Ibid.

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Funding Forest Businesses

Vance Russell

Chapter updated to include 2025q1 grant awards

Findings

To better understand the geographic distribution, success rates, and funding gaps/needs, the Forest Business Alliance examined CAL FIRE's Business and Workforce Development Grants from the first quarter of 2022 through the first quarter of 2025.¹ The analysis of CAL FIRE's Business and Workforce Development Grants Program found the following:

-  **Program Funding.** Funding requests far outstrip the grant funds awarded.² A steady linear increase in funds requested from 2022-2025 has been counterbalanced by a decline in total grants funded. For Q1 2025, approximately \$85.7 million in requests from more than 100 proposed projects were made, with \$5 million awarded. To date, the total program requests since 2022 are just north of \$457 million, with \$105.7 million funded.
-  **Top 10 Counties.** The top 10 grant-receiving counties in order were Tuolumne, Shasta, Humboldt, Plumas, Tulare, Yolo, Placer, Siskiyou, Fresno, and Sonoma Counties.
-  **No Funding.** The counties that have not yet received funds are Colusa, Del Norte, Imperial, Inyo, Kern, Kings, Merced, and Santa Barbara. Imperial is the sole county not making a funding request and is unlikely to do so, given its landscape dominated by desert and agriculture.
-  **Regions.** As of March 2024, the largest portion of grants was awarded to the Sierra Cascade (61.6%) and North Coast (31.6%) regions. The Central Coast (5.2%) and Southern California (1.6%) were far behind, although both regions' requests have increased substantially during the last two grant solicitations.
-  **Success Rates.** The counties with the highest grant success rate (proportion awarded/requested) were Stanislaus, Sutter, Solano, Alpine, Siskiyou, Lake, Modoc, Mariposa, Santa Cruz, and San Benito counties. However, a bivariate analysis that simultaneously examined requested and awarded amounts showed that the most successful counties (those with high amounts awarded and low amounts requested) were Alpine, San Francisco, Sacramento, San Joaquin, Sutter, and Ventura Counties.
-  **Analysis & Location.** The applicants' project locations need to be more precise to increase the accuracy of the analysis. Accurate location data for applicants and grantees would also help identify counties and regions with the most need, such as disadvantaged communities.

Background

Prior to European settlement, fires were commonly utilized throughout California by Tribal communities as a tool for managing food, game, disease, and community safety. Approximately 4.4 million acres burned annually in California before 1800 (Stephens et al. 2007). This acreage is far more than in recent years, but likely differs in the total acres burned at high severity. Fast forward—fire suppression, severe drought, and climate change have created forest ecosystems well outside their natural conditions, making them prone to severe wildfires and negatively affecting habitats and communities.

This century's increase in large fires has underscored the urgent need for forest health projects. These projects aim to thin dense forest stands and reintroduce fire into forested ecosystems (Figure 1). However, they have also created a new challenge—excess wood and biomass that pose additional wildfire risks. California is in dire need of more infrastructure and workforce capacity to manage this situation effectively. The USDA Forest Service's Wood Innovations Grant Program encourages development of wood products and processing at local to regional scales. At the state level, CAL FIRE created the Wood Products and Bioenergy Program with the first grants

¹New data and analysis will be added during subsequent grant rounds.

²Forest product infrastructure development is a state priority for wildfire mitigation. Sustainable infrastructure to process wood biomass in California and workforce development are critical elements of the [Roadmap to a Million Acres](#).

awarded in 2022. Both programs aim to fund projects that process woody biomass and build capacity for work in the woods.

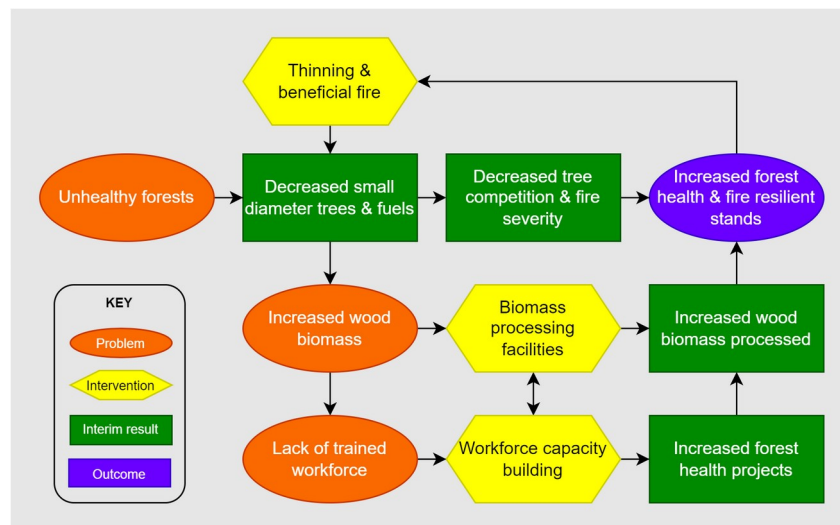


Figure 1: Simplified forest health theory of change.

Methods

The funding analysis examined awards by county from 2022-2025 for the CAL FIRE Business and Workforce Development Grant Program (Figure 2). The funding data is available on CAL FIRE's Wood Products & Bioenergy website. We extracted the requested and awarded data from the website and added them to an Excel file. We added a sub-categorization of projects by type.

To conduct the analysis, we excluded statewide grants since they could not be geographically assigned. For projects covering multiple counties, we averaged the total requests and awards across each county identified for the project. The data was compiled in Excel and then joined to a county spatial layer (California Tiger Census) and a vegetation layer (Calveg) to create the spatial database.

The spatial analysis was completed using ArcGIS Pro. See [Local Bivariate Relationship](#) for an explanation of the bivariate analysis tool. Graphs were produced within Excel using power pivot and pivot charts.

Analysis Awarded

Total amounts awarded by county are shown in Figure 2. The darker green counties indicate a higher amount awarded whereas purples and blues indicate lower amounts.

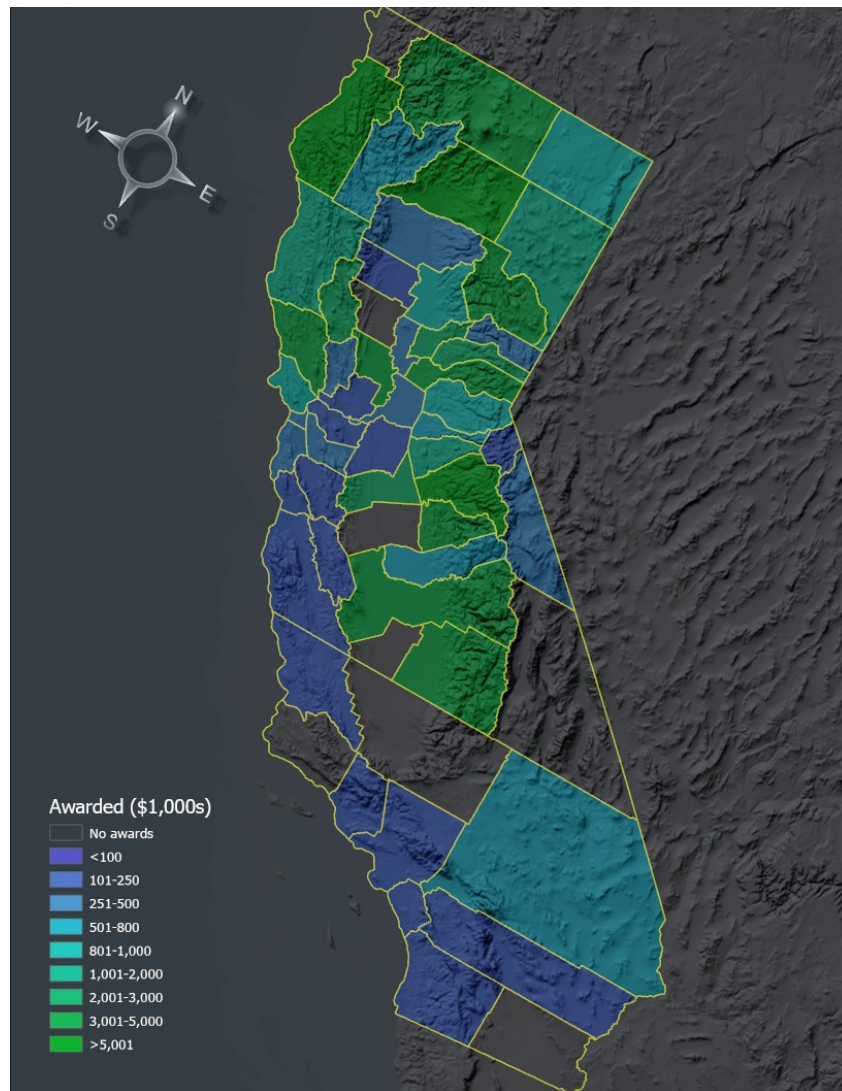


Figure 2: Awarded grants by California County from 2022-25. Numbers in thousands \$USD.

Norcal

Not surprisingly, the highest number of grant requests for the Business and Workforce Development Grant Program is in the northern part of the state, with most from the Sierra Nevada (Figure 3). Surprisingly, some agriculturally dominant counties rank highly, e.g., Yolo County, partly because large bioenergy projects were funded there.

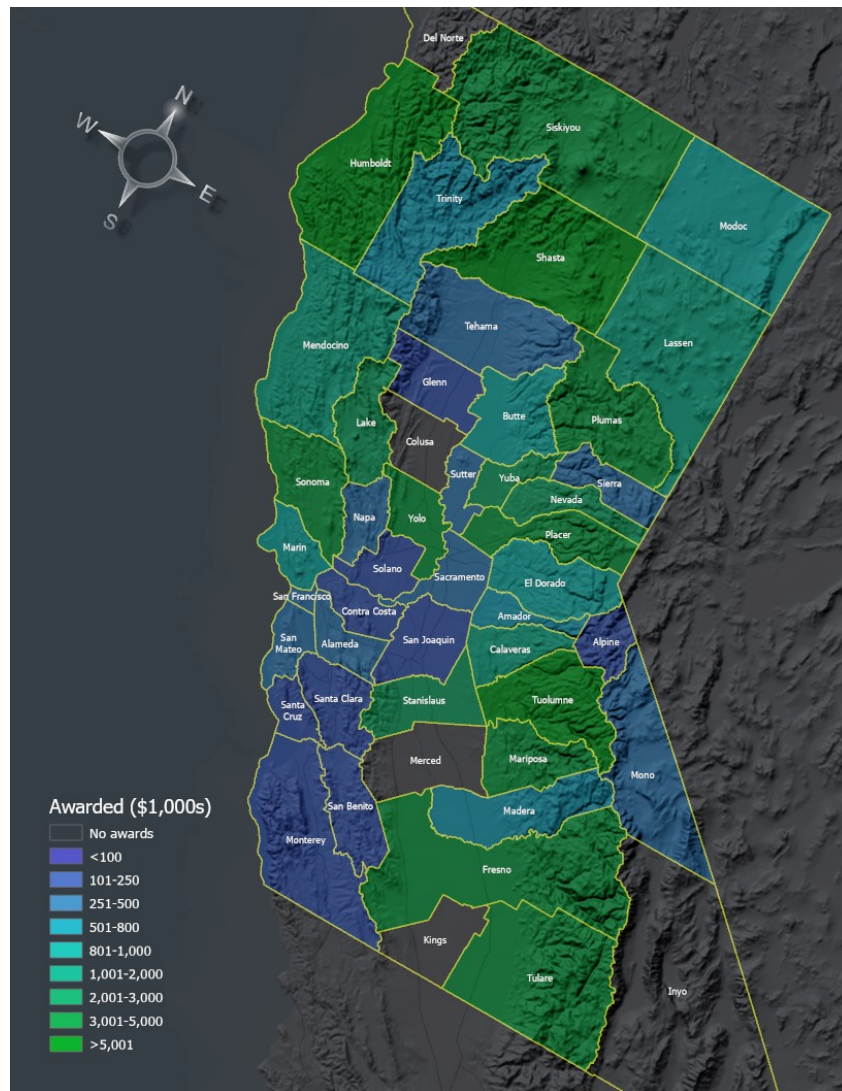


Figure 3: Business & Workforce Development grant funding in Northern California.

Socal

Much of Southern California is characterized by chaparral and desert landscapes, and fire mitigation in this region focuses on preventing ignition sources rather than thinning forests or processing wood products. As a result, there are fewer proposals, and many of the requests are focused on workforce development. Before 2024, only two counties in Southern California had projects with funding: San Luis Obispo and Ventura. However, this changed substantively in 2024 when all but Santa Barbara, Kern, and Imperial Counties had received funding (Figure 4). Grant from the region substantially increased for requests in the last grant round of 2025.

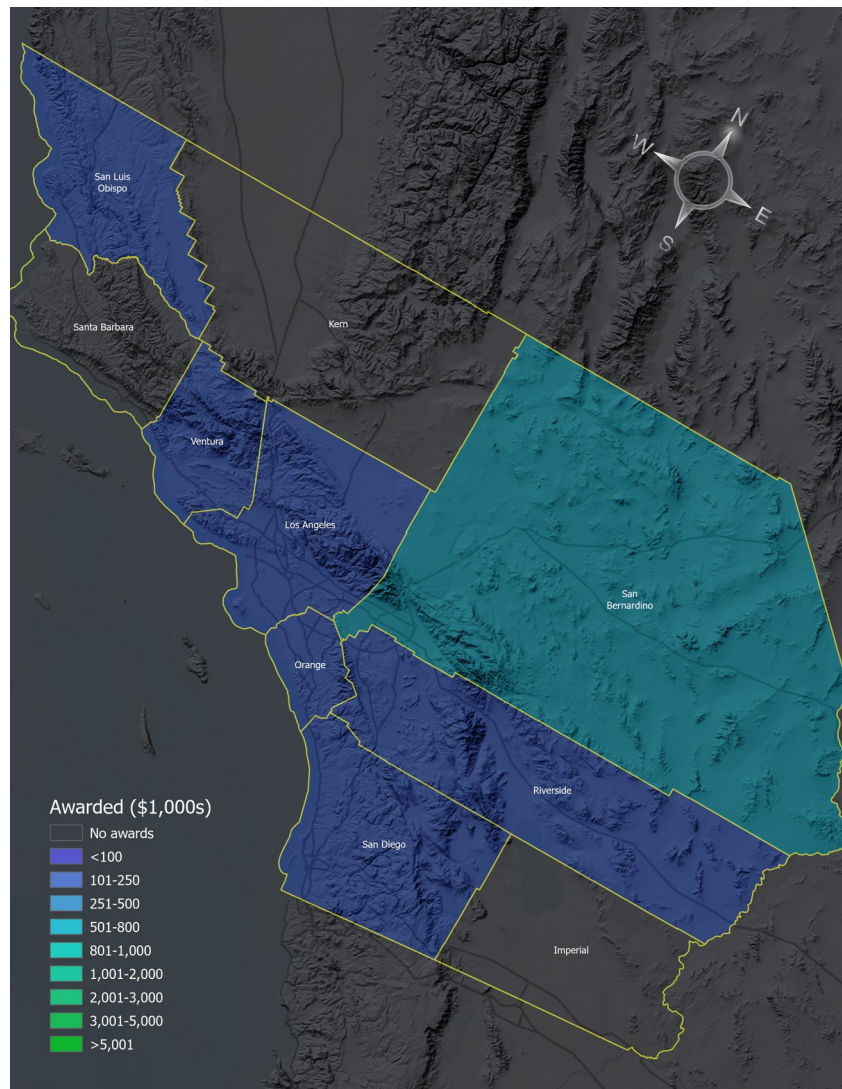


Figure 4: Grant funding in Southern California.

Bivariate

A bivariate analysis, which queries the data by total requests and awards, paints a slightly different picture than the awards analysis (Figure 5). Where the success rate is the relationship between funded and requested grants, the lightest shades tend to indicate low success rates, whereas the light blue to dark purple indicates counties with high grant success rates.

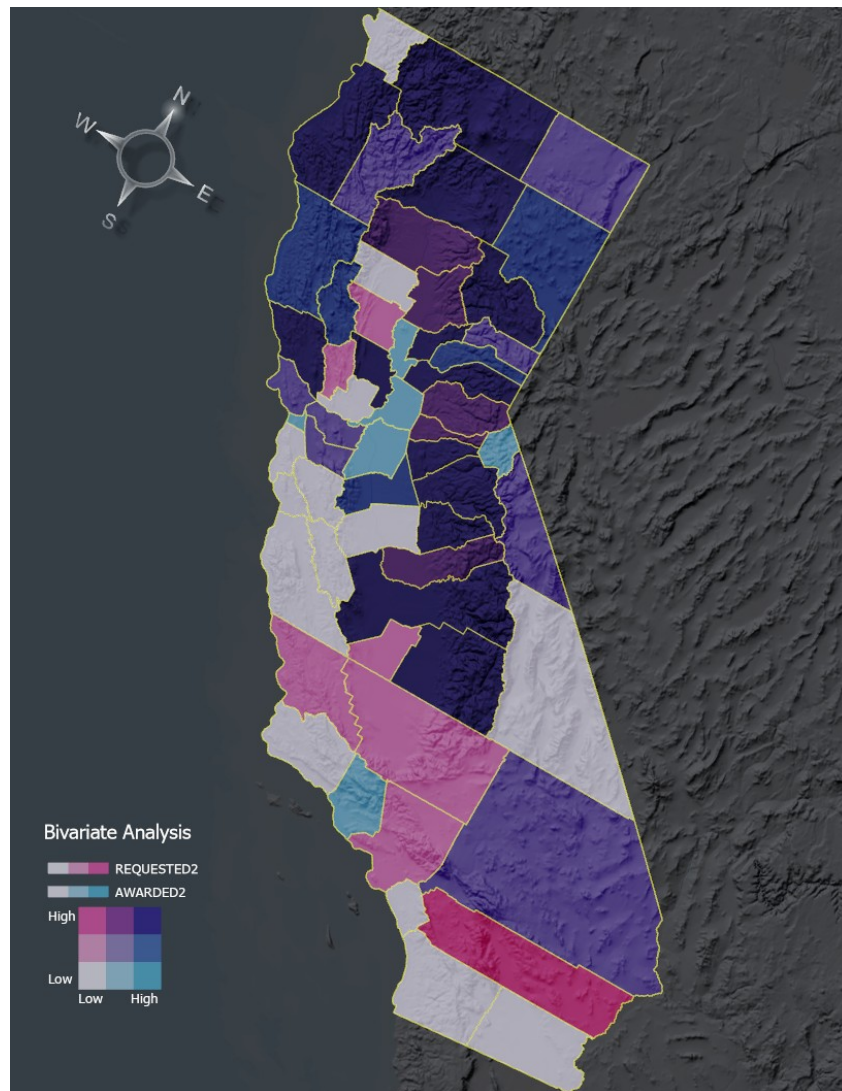


Figure 5: Bivariate analysis comparing amounts of requested to awarded grants.

Figure 5 reveals some Southern California counties have requested grants but have not been successful (e.g., San Luis Obispo, Kern, Orange, Riverside). Ventura County has a high success rate, with a high percentage of awards requested (shown in light blue). In the rest of the state, Sacramento, San Francisco, Sutter, and Alpine Counties show similar high success rates. In contrast, counties such as Tulare, Plumas, and Shasta have received a substantial amount of funding but have also submitted numerous requests.

Gaps

The total funding requests and awards by county reveal the divide between the northern and southern portions of the state (Figure Figure 6). The largest number of grants requested and awarded was in the Sierra Cascade Region. Nevertheless, the totals do not fully explain the project quality in the region. For instance, Inyo, Kings, and Merced have not funded any projects. The Forest Business Alliance addresses the lack of capacity in each entity's ability to apply for grants to strengthen sustainable forest businesses and process more wood, ultimately increasing forest health across the state.

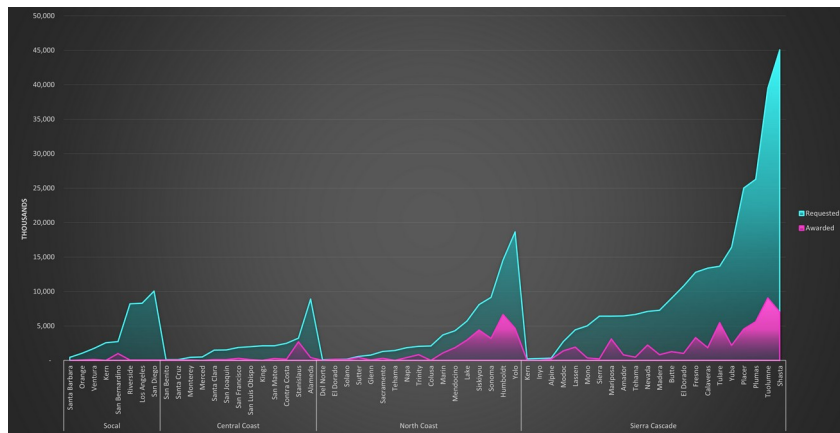


Figure 6: Stacked bar chart of requested and awarded grants by Wildfire Task Force Region and county.

Funding by project type showed the most funding for workforce development training, biomass, transportation, and equipment projects (Figure 7). Business development, marketing, and thinning projects were the least funded project types.

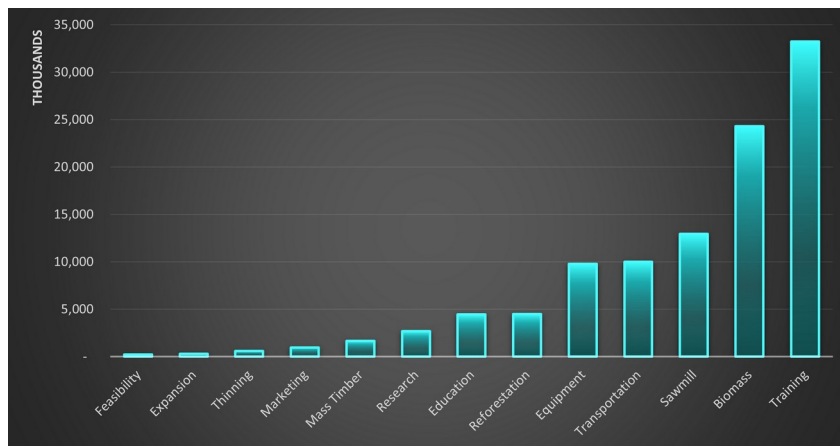


Figure 7: Awarded grants by project type.

The number of unfunded projects by county reveals interesting patterns (Figure 8). Some counties with the most unsuccessful applications are in the Sierra Cascade Region, e.g., Tuolumne, Shasta, and Plumas. Counties with low success rates submitted fewer proposals, e.g., Inyo, Kings, and Alpine.

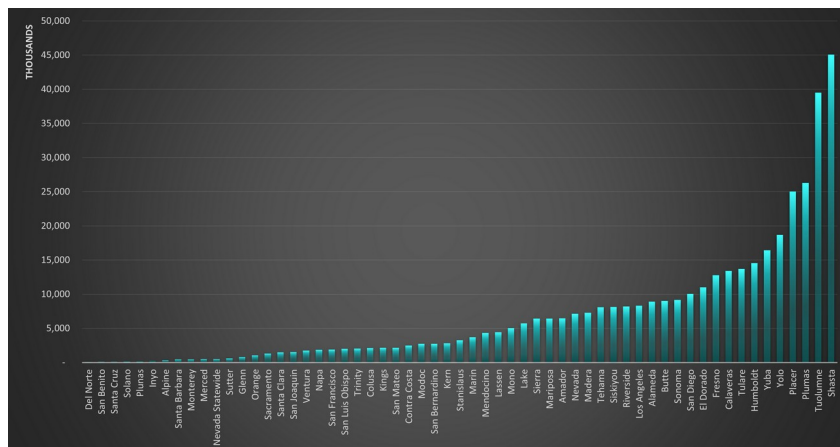


Figure 8: Unfunded requests by California County.

Temporal

The Business and Workforce Development Program funding changed substantially from 2022. Figure 9 shows the requested and awarded grants by year and region, with larger amounts requested from the Sierra and Central Coast. However, the Central Coast and Southern California have started to catch up in requests. SoCal has not caught up in successful grants, however.

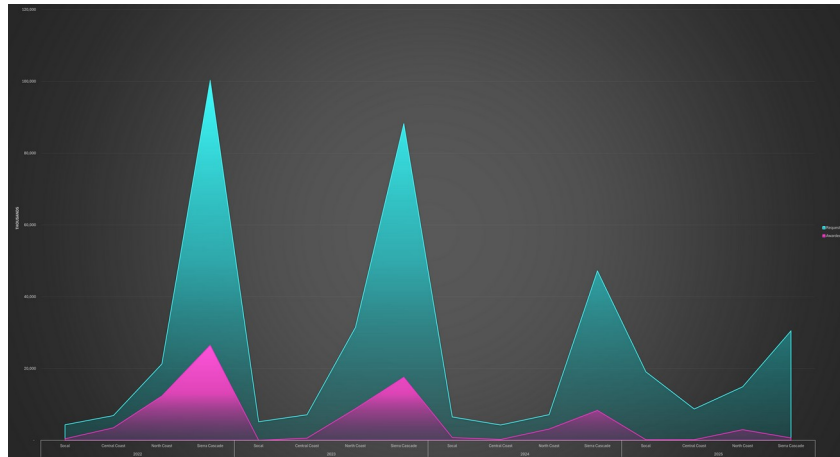


Figure 9: Requested and awarded grants by year and region.

Next

The Forest Business Alliance will continue to add new projects to the Business and Workforce Development funding analysis database in response to new grant solicitations. We recommend the following:

- **Refine project location.** More accurate project locations, e.g., latitude/longitude points, should be added to reflect where the funding is directed and to examine the relationship between forest health and wood products businesses. We will include new analyses, such as those from new projects funded through subsequent grant rounds. How do the funding success rates change geographically and temporally?
- **Additional analysis.** More analysis is needed to determine the reasons for the county and regional differences. Is it organizational and business capacity? Is it related to the capabilities to create strong projects and proposals? Could it be something more pervasive, such as regional economic differences or the availability of institutional support for potential applicants?
- **Increased funding.** Given the need for wildfire mitigation in the state, the bottleneck in processing biomass in a sustainable manner, and the mismatch between requested and funded businesses, we highly recommend that the state consider increasing funding for this valuable program.

Funding

The CAL FIRE Business and Workforce Development Grant Program and Yuba Water Agency provide funding for the Forest Business Alliance. [CAL FIRE's Wood Products and Bioenergy Program](#) manages the BWD grants and works to maintain and enhance California's wood products infrastructure to support healthy, resilient forests and the people and ecosystems that depend on them.

[The Forest Business Alliance](#) provides technical assistance, workshops, and a peer-learning network to increase local and regional capacity for California wood products and forest health. Funding for this project is provided by CAL FIRE's [Business and Workforce Development Grants](#). Please don't hesitate to contact us at forestbusinessalliance@gmail.com with any questions or feedback.



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Scaling

Vance Russell

Takeaways

1. **Tailoring strategies.** Off-the-shelf solutions do not exist, especially in the forest health arena. Tailored strategies for scaling are therefore required, but bespoke solutions make scaling challenging, particularly if they have to be adjusted for each situation, geography, or organizational component.
2. **Pilots may not work.** Sometimes referred to as the “pilot and pray” approach. In many cases, we may be running out of time, funding, and resources to conduct pilots. However, if you have to pilot a project, try it at a small scale, with clear hypotheses and metrics set up to test effectiveness.
3. **Measuring outcomes.** Create monitoring and measurement systems that measure outcomes or impacts of forest health projects and your company’s success. See the [Monitoring](#) chapter for more details on creating a system to measure your business success.
4. **Workforce.** Finding qualified employees is a major challenge for the private and public sectors. Registered professional foresters (RPFs) are in short supply, and knowledgeable mill operators are difficult to find and train. Even assembling qualified thinning or prescribed fire crews led by experienced individuals can be difficult. Challenges are increased for nonprofits that offer lower salaries than the private sector.
5. **Leadership.** From an ownership perspective, company leadership can be tricky, e.g., letting go as the company founder when the business expands. Depending on organizational needs, leadership may be needed in different locations or over time.

Background

Surprisingly, little is written on scaling nonprofits. Overwhelmingly, a lot is written on scaling businesses; much of it is breathless commentary about tech companies moving from a single founder to 1,000s of employees and how great the fireman’s pole at the SOMA office in San Francisco is. Few to none of this *oeuvre* is linked to forest health, with most articles focused on scaling tech startups and most exploring scaling nuts and bolts.

Emotions

Molly Graham, from her experience of scaling at Google, Facebook, and Quip, takes a unique scaling approach, examining the people and emotional side of scaling, noting that you should not try avoiding or ignoring the emotions that couple with the scaling tornado, but acknowledging they’re normal and any other organization will be going through similar circumstances Graham (2024). She says that emotional acknowledgment, which she likens to making friends with the monster chewing on your leg, is only half the battle; the other half is how you respond (Figure 1).



Figure 1: Acknowledging and responding to the monster chewing on your leg is critical for managing scaling. Graphic credit: Graham (2024).

Considering emotions in scaling is crucial for managing the workforce and also an important part of scaling leadership. Let's look at other approaches from the conservation and agricultural fields.

Up/out/deep

In the conservation arena, Salafsky *et al.* (2021) adapted a framework created by Moore *et al.* (2015) for scaling conservation nonprofits. Based on their literature review, there are three scaling approaches:

1. **Scaling out** involves replicating the initial pilot with three options: 1. expanding the scope, 2. replicating pilots within a program, and 3. promoting innovation diffusion by capturing and communicating what you've learned and getting other organizations to adopt your strategies.
2. **Scaling up**, e.g., policy change, takes a systems-level approach to leverage scale across a greater area rather than testing and replicating pilots.
3. **Scaling deep** is a behavioral change approach that changes the underlying values of the actors in a system. For example, it might involve building a stewardship ethic among landowners or creating and communicating stories that convey the needed values to key audiences. It's best to take a careful, measured approach to scaling deep. In many cases, this type of strategy takes a lot of time and requires many touch points and well-thought-out strategies to succeed. Simply running radio spots to bolster messaging isn't enough to change behavior.

Scaling out is common in conservation but has several downsides, including high risk and cost. Piloting may ultimately lead to what many conservation projects follow the 'pilot and pray' approach Salafsky & Margoluis (2021). In other words, implementing conservation actions based on effectiveness while ignoring scalability risks Pienkowski *et al.* (2024). A bottom line reality is that we may frequently be beyond the pilot stage for forest or biodiversity conservation due to the advanced state of wildfires, climate change, and habitat loss. Pienkowski *et al.* (2024) says the inverse can be true, e.g., promoting scalability over effectiveness, and that effectiveness can change with scale, possibly requiring different methods to measure outcomes. For example, voluntary carbon markets have failed to deliver benefits due to inadequate methods to measure additionality West *et al.* (2023).

Farmer-2-farmer

Compared to agriculture, conservation falls far behind in understanding the factors related to adoption and scale. In his seminal book on farmer-to-farmer extension called *Two Ears of Corn*, Roland Bunch describes scaling agricultural extension, starting small and spreads by diffusion Bunch (1982). One of the main components of this approach begins with small-scale experimentation. Farmers drive experimentation at this level on tiny, ~50 m², plots, creating a low-risk approach. Small-scale experimentation has several advantages at the programmatic level since it can reach the poorest farmers due to its low-risk, low-cost approach.

As more farmers do more experiments and successful strategies are chosen, the diffusion of the program spreads rapidly as farmers adopt cost-effective farming practices. The approach also has the advantage of being self-sustaining and does not need technical assistance from outside experts since farmers learn directly from one another; in other words, the program creates and turns its flywheel Collins (2005). Like the scaling deep approach mentioned above, farmer-to-farmer approaches can be scaled as movements and connected to social movements for food sovereignty Brescia (2017).

Brescia (2017) offers a similar scaling framework to Salafsky *et al.* (2021):

1. **Depth.** Groups of farmers innovate on their farms.
2. **Breadth.** Known as horizontal scaling and can be achieved when practices are spread across many households and communities.
3. **Verticality.** Comes into play when enabling farming practices across networks and movements and linking community farmers to markets and supportive policies.

Planned & scaled funding

Funding is an important factor in scaling not covered in the agricultural examples above. In the check your scaling assumptions Salafsky *et al.* (2021) provides an excellent example of developing realistic budgets based on strategic plan goals and costing those out over time. They show that to reach a seemingly realistic but ambitious goal of 500,000 tons of CO₂e sequestered, a hypothetical organization will have to carry out 250 projects over a decade. This is when s*%t gets real! They then create a financial model to determine total projects, costs, tons of carbon sequestered, and cost/ton. The upfront work demonstrates the value of testing assumptions made to carry out a plan Salafsky *et al.* (2021). A similar analysis could be completed simultaneously, focusing on staffing and workforce needed to carry out the plan.

Forest Businesses

Scaling businesses is a double-edged sword: while it is a well-trodden path with numerous success stories, the pressure to scale begins immediately, accompanied by expectations for profit and system-wide success. Factors business owners may want to consider when planning to scale include

- **Finance.** As a business scales, Different timings, amounts, and for different items may occur, further complicated by changes in markets and other external factors. Financing from traditional institutions for new products and approaches may take longer to develop but should become easier as more forest health businesses come online with successful track records.
- **Funding.** A business mantra for funding could be to use government grants to test and loans to scale. Government grants can be used to test new ideas, practices, and equipment and act as a powerful innovation and test phase before scaling. Phased properly, along with purposeful testing, grants can allow pilots to be rapidly scaled. Following successful test implementation, loans can scale landscape implementation following the proven track record developed with agency grants. As highlighted in [Program Design](#), use your business or strategic plan to prioritize the grants you pursue rather than allowing grant programs to change your priorities and vision.
- **Leadership/ownership.** Company leadership can be tricky from an ownership/ideas perspective, e.g., letting go as the business expands. New leadership may be needed to manage satellite businesses and hire the right leadership to maintain corporate culture. Company founders may be used to overseeing all aspects of project and company development, so delegating those tasks and creating the structure to do it effectively becomes more critical as a company grows. Succession as a company matures carries the same needs and cannot be overlooked, particularly during rapid growth. Similarly, over-dependence

on the vision of the original leader/founder can hamper a company's ability to scale and be successful beyond the original business bounds.

- **Markets.** Many forest product markets are new, making them difficult to create. Creating a cooperative across multiple businesses to weather product startups may help, or entering new markets slowly, deliberately, and at small scales until they are established or you have consistent buyers is another. Another way may be to diversify your product lines with a mix of traditional items to create a core revenue source and new products to test their selling power and establishment.
- **Quality control.** Maintaining quality control with franchises can be difficult, especially when the original product artisans can no longer oversee every aspect of production. Training new staff for satellite production facilities should be done using the scaling procedure and given enough time. Quality control can also be compromised when rapid production increases are required. Keeping pace with this change is difficult to predict but must be factored into expansion or scaling plans.
- **Maintenance.** If you're focused on forest health and restoration, maintaining previously treated sites should be built into your scale model with workforce, transportation, equipment use, and distance to sites for processing wood all considered. Maintaining equipment, vehicles, and other hardware in your business depends on what needs to be built into your costs and how you scale and have the right people to maintain needed equipment schedules.
- **Sustainability.** How sustainable is your business as it expands? Are there controls and principles in place to maintain sustainability? Creating and maintaining a focus on a triple bottom line, measuring your company's performance based on profit, people, and planet (or environmental factors), building in B-corp certification, or giving back to the community can be significant factors. At another level, how do all of the factors in this section help your business weather change over time, and how do you measure sustainability or resilience to changes in supply, production, and markets?
- **Measuring outcomes.** Create monitoring and measurement systems that measure the outcomes or impacts of forest health projects and your company's success.

Challenges

Rapid scaling is a huge challenge, particularly for nonprofit organizations that are often unable to scale rapidly due to funding, workforce, and governance issues. These issues seem overwhelming, but each piece is not insurmountable and may frequently have simple solutions. Patterns of challenges for scaling include

- **Workforce.** Finding qualified employees is a major challenge for the private and public sectors. RPFs are in short supply, and knowledgeable mill operators are difficult to find and train. Putting together qualified thinning or prescribed fire crews led by experienced individuals is challenging. Challenges are increased for nonprofits that offer lower salaries than the private sector.
- **Chicken and egg finance.** Banks are no longer experienced in-house with the due diligence required to review loans for forest health and infrastructure projects. To compensate, they ask businesses to return once the business is functioning. This chicken and egg situation, akin to not being able to apply for a job out of college w/o experience, is trying for new businesses or existing businesses trialing new technologies. This challenge may decline over time but certainly does not help accelerate the pace and scale of forest health treatments at the present moment.
- **Funding & scale.** Funders often ask for scalable projects but don't provide the necessary funding to scale once a project is successful. Also, public investment through grants, which is fantastic for trialing new and innovative programs or equipment, is not being leveraged with private funding.
- **Nonprofits scaling pace.** Nonprofits typically do not scale quickly. It's impossible to scale when you fund many projects based on a reimbursable format, such as funding from most state and federal agencies.
- **Agency agreements impede scale.** Currently, the slow speed of executing agency agreements—some may take upwards of one year to follow grant awards—significantly impedes the pace and scale of forest health implementation. The cost and speed of environmental compliance are similar issues, putting the brakes on project implementation.
- **NIMBY.** Neighbors may oppose projects despite the need for biomass utilization, forest products, and mills. For example, Alpenglow Timber in Truckee, California, in late 2024, had a NIMBY challenge even though some of the project opponents were on biomass promotion projects or boards!
- **Project size ≠ need.** Coordinating across multiple actors in a region can be particularly challenging and take a lot of time, time that some organizations do not have when starting new entities. For example, projects in the California Wildfire Task Force tracking dashboard may look small compared to the actual goal of treating 1 million acres throughout the state.

- **One size doesn't fit all.** This makes for a conundrum that prevents scaling since every effort to scale requires bespoke changes in policy, funding, workforce, behavioral, and institutional change Churches (2023). Furthermore, some companies may be content to scale to a certain size; others may wish to grow further. This right size can be factored into planning by the organization's board, but bottoms-up ownership of this growth also needs attention, not just top-down direction from the board.
- **Effectiveness & scalability.** Evidence of scaling effectiveness, readiness, and risk evaluation can be lacking in projects. The actual total impact is not dependent on scalability alone, it is the sum of effectiveness and scale Pienkowski *et al.* (2024).

Recommendations

Overall, strategies need to be tailored. Considerations to incorporate scale when developing your business, project, and programs for forest health:

- **Scale from the outset.** Consider scaling from the start of your program design.
- **Calculate scale costs.** Rough out the costs of scaling your work and assess feasibility.
- **Emotional scale.** Be aware of the emotional cost of scale to you as a business owner or nonprofit manager and its impact on your staff.
- **Replication & adoption.** Replication and adoption are moving targets and change as you scale.
- **Effectiveness.** Measuring success becomes more complicated with scale but infinitely more valuable for learning and determining what is working or not.

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Grant Management

Christopher Sacco, Vance Russell, Kelly Kahling

Takeaways

1. Grant management is critical to project and organizational success, yet doing it well is often overlooked or given a lower priority.
2. Consistency with funders, whether they're public, foundation, or private, is of utmost importance. Don't assign grant management to interns, new staff, or temporary staff.
3. Get to know your program officer or grant manager and make their life easy. See number 2, have the same person manage the grant for its entirety.
4. Create a strong, auditable, and accurate financial accounting system that also automates reporting or is flexible for each funder to create reports from your P&L ledger.
5. Make reporting do double duty. At its worst, reporting can be drudgery. Change that and make reports effective communication tools for your project and organizational successes. Build reporting into your organizational learning and evaluation cycles. Some organizations use their annual report for both public outreach and reporting to donors.

Budget

"A budget is telling your money where to go instead of wondering where it went."

John Maxwell

Preparing a good budget during the proposal stage makes grant administration easier. A solid budget prevents surprises and serves as a roadmap for managing and tracking resources effectively.

Some common reasons why good organizations and smart people prepare bad budgets include:

- Lack of coordination/disconnect between technical/program and budget/finance (so cost and tech don't align)
- Costs that are not based on historical or real data
- Most time spent on design/technical, so not enough time allocated for budget
- Activities outlined during the design phase cost more than the money available

After you win a grant, budget tracking is essential:

- Ensures Funds Are Spent as Planned: Aligns expenditures with the approved grant budget and timeline.
- Monitors Progress & Performance: Helps track spending against milestones and deliverables.
- Identifies Issues Early: Flags overspending, underspending, or misallocations before they become problems.
- Supports Accurate Reporting: Provides documentation for financial reports, audits, and funder updates.
- Enables Informed Decisions: Real-time tracking supports smart adjustments and resource reallocation if needed.

Automate Expense Tracking

Automate expense tracking as much as possible. There are numerous apps available that enable employees to track expenses and upload receipts. These can be fed directly into your accounting system.

Match

Match tracking is essential for meeting grant obligations and can be critical for leveraging future grants:

- Meets Grant Obligations: Many grants require matching contributions (cash or in-kind); accurate tracking ensures compliance.
- Provides Verifiable Documentation: Track source, type (cash/in-kind), value, and date of each match with proper records.
- Supports Audit-Ready Reporting: Matching funds must be documented and reportable, just like grant expenditures.
- Demonstrates Stakeholder Commitment: Shows funders that your organization and partners are invested in the project.
- Prevents Disqualification: Incomplete or inaccurate matching fund records can jeopardize current or future funding.

Match Plan

Create a plan to meet the match requirements and track your progress from the start.

Planning

Just as a good budget is connected to financial management and reporting, a solid implementation plan is required for tracking progress toward your project's goals and objectives. Although you should have fleshed out activities when designing your project and creating its budget, the proposal you submitted often lacks the level of detail required by project managers for implementation.

Once funding is obtained, detailed implementation planning takes place to produce updated schedules, plans, targets, and systems that have sufficient detail to permit effective project implementation. Some organizations call this start-up planning. Detailed implementation planning is closely tied to monitoring, evaluation, and learning.

DIP

Why do detailed implementation planning (DIP)? Detailed implementation planning helps ensure that a project's contractual obligations are fulfilled. These include the following:

- Conducting project activities that lead to the delivery of outputs that meet quality standards, are on time, and within budget;
- Using and managing project resources in conformity with budgets and accounting standards and rules; and
- Complying with any relevant conditions of the Award Agreement.

DIPs are usually prepared after a proposal is approved and funded, but before implementation begins.. DIPs may be undertaken on an annual basis, for the life of the project, or both. If completed for the life of the project, the DIP is still revised and updated annually. DIPs are referred to as annual work plans.

Proposals are not DIPs

Remember: Your project proposal is not a DIP, because it does not have the required level of detail to use as a plan for implementation. Even the best M&E plan in a well-done proposal needs to be operationalized during detailed implementation planning. Operationalizing means making the plan operational, i.e., providing sufficient detail so that it can be implemented. However, DIPs do not start with a blank page. A proposal provides the basis for a strong DIP through its Proframe, M&E plan, Activity Schedule, organizational structure and staffing plans, budget, and other elements. Many of the steps in detailed

Implementation planning involve reviewing, updating, correcting, and further specifying elements already contained in the proposal.

If you use FBA's templates and guidance to design your project, you will have at least the beginnings of DIP and a project tracking system. FBA created the ProFrame/DIP worksheet specifically to identify CAL FIRE metrics and set associated targets required for the application.

Monitoring

Design teams, especially those creating complex, multi-year projects, should consider drafting a MEL plan for their use that also responds to CAL FIRE's information needs. Chapter 4 Monitoring provides a primer to help you create a MEL plan. In developing a MEL plan, you would create specific, measurable, achievable, relevant, and time-bound (SMART) indicators for all your goals, objectives, intermediate results, and outputs.

The e-Civis Grant Portal Goals worksheet, which is used to input project metrics and targets, does not require fully articulated indicators. Nevertheless, it is worthwhile to develop such indicators, as well as the rest of the MEL plan, to facilitate your monitoring and evaluation work.

Reporting

All funding organizations require activity (progress) and financial reports. These reports must be completed accurately and promptly. This helps you as much as it allows your grant organization by keeping you on schedule, on track with reporting, ensuring transparency with your funding agency, maintaining compliance, and supporting future funding opportunities.

CAL FIRE

CAL FIRE usually requires an activity report with every invoice, and prefers them to arrive together. This is where you will track your project milestones and deliverables. You will have the opportunity to describe your accomplishments, challenges, and highlight any adjustments or improvements you plan to implement. It is a good time to review your approved scope of work and make sure your activities are aligned with your grant deliverables. Activity Reports are submitted on a regular schedule (CAL FIRE requires at least one per quarter, not more than one per month).

Grant Invoicing & Reporting Tips

- Get organized!
 - Use standardized templates and automate portions of your report for accuracy and ease of replication.
 - Always communicate with your grant manager about any issues.
 - Make sure you have backups, especially for financial information
 - Use reporting to benefit your business/organization. **Don't make reporting drudgery!** Take advantage of it to build organizational learning and communication assets. You decide on the reporting format. Sometimes, donors may unnecessarily request a lot or in a challenging format. Work with them constructively to reduce friction, automate processes, and achieve productive outcomes.

Financial

Financial reporting is where you lay out your costs associated with the activities performed during the reporting period. You will provide detailed expenditures by budget category. You would also include any matching contributions you may have that were recorded during the Reporting Period. It is important to produce backup documentation (e.g., receipts, timesheets, and vendor invoices). Financial reports must reconcile with your

internal accounting records. Your Financial Report is a summary of what your costs were for your Reporting Period. Don't forget to stay on track with your grant timeline and budget.

Automation

There are many software packages for managing and automating financial tracking and reporting. Most are focused on small to medium-sized businesses and may offer reporting templates, but they are not typically relevant to forest product businesses or nonprofits.

[Miradi](#) is an exception to the general financial accounting software. It is available online or as a desktop version and nicely integrates projects from situation assessment, theory of change, results chains, progress tracking, to reporting. It does have a subscription plan of \$350/year (2025), more if you want to customize report templates. Still, the community, connection to many foundations and nonprofits that adopted the [Conservation Standards](#), and integration of the entire project make it a worthwhile investment if you choose to operate in that system. It is geared more towards nonprofits, however, so it may not be as useful for small business owners.

Try the Python automation if you're comfortable using a command-line interface in a terminal or prefer a DIY approach. Use the executable example if you want similar automation but using a graphical interface.

Python Example

We offer a simple example that requires some setup time but automates your reporting in a customizable format. Similar to any recipe, there are some *mise en place* steps to undertake before getting to the fun stuff, especially if you've never coded before. Don't worry, none of it is difficult, and once you set it up and it works, you can step outside your office, beat your chest, and yell, "I am a coding superstar!". Well, maybe don't yell that. Here are the setup steps¹.

1. Install [conda](#). Decide if you want the lightweight command interface, miniconda, or the desktop-based Anaconda.
2. Open a terminal. In Windows, press the Windows key and type 'terminal,' then open the Terminal app or Windows PowerShell. You should see the following:

```
(base) C:\Users\yourusername>
```

1. Create a conda virtual environment. Never install packages into your base environment to avoid incompatibilities, which can affect your base (that you can't delete) and make troubleshooting easier. Decide on your environment name; let's call it 'auto' (use a short name, five letters or less), in this case. Type the following command in the terminal and hit enter:

```
(base) C:\Users\yourusername> conda create -n auto python
```

The terminal will start running, and when it says **Proceed ([y]/n)?** hit enter or y on your keyboard. Conda will download the necessary files and provide instructions for activating or deactivating the new env.

1. Check installation and activate the auto virtual environment.

```
(base) C:\Users\yourusername> conda env list
```

You should see a list of the conda environments installed, including auto, and their respective locations. The active env should be the base and have an asterisk next to it. Now activate auto:

```
(base) C:\Users\yourusername> conda activate auto
```

¹Detailed steps for virtual environment and Visual Studio Code setup can be found in the 1st two appendices of [Free & Open Source Geospatial Tools](#).

You should see the following:

```
(auto) C:\Users\yourusername>
```

1. Install autorpt:

```
(auto) C:\Users\yourusername> pip install autorpt
```

The install will download and install. You can verify the installation by typing `conda list`, and it will appear alongside the other installed packages.

1. Run autorpt:

- Download the budget.xlsx and content.md files from <https://github.com/VRConservation/autorpt> and store them in a folder of your choosing.
 - Open the folder, right-click inside it, and select “Open in Terminal”.
 - Activate the virtual env by running `conda activate auto` or use whatever name you’ve decided.
 - On the command line, type `autorpt`.
 - `autorpt` will run the script, you’ll see progress in the terminal, and a report in Word will be produced.
 - The bugs are still being worked out, but you will be able to create a PDF of the report by running `autorpt --pdf` soon.
 - Modify content.md with new information each time you need to create a report. Each time you run `autorpt` a new version will be made so you can edit drafts and not overwrite previous versions.
 - This is an example only, and we get that you could copy and paste in Word and Excel to create a report, but this eliminates those steps and lets you focus on the report content rather than production. You can also modify the report content with new sections or rename the sections according to your reporting requirements.

Python Executable AVAILABLE SOON!

Alternatively, if you’re not comfortable using a terminal and command-line interface, and would rather click, you can download `autorpt.exe`, which opens a user interface with Word and PDF document options (Figure 1). Download [gui.exe](#). Make sure the .exe file is stored in a file with the content.md and budget.xlsx, and the interface will generate the reports for you when pressing the buttons.

PLEASE NOTE THIS IS NOT YET FUNCTIONAL BUT WILL BE AVAILABLE SOON

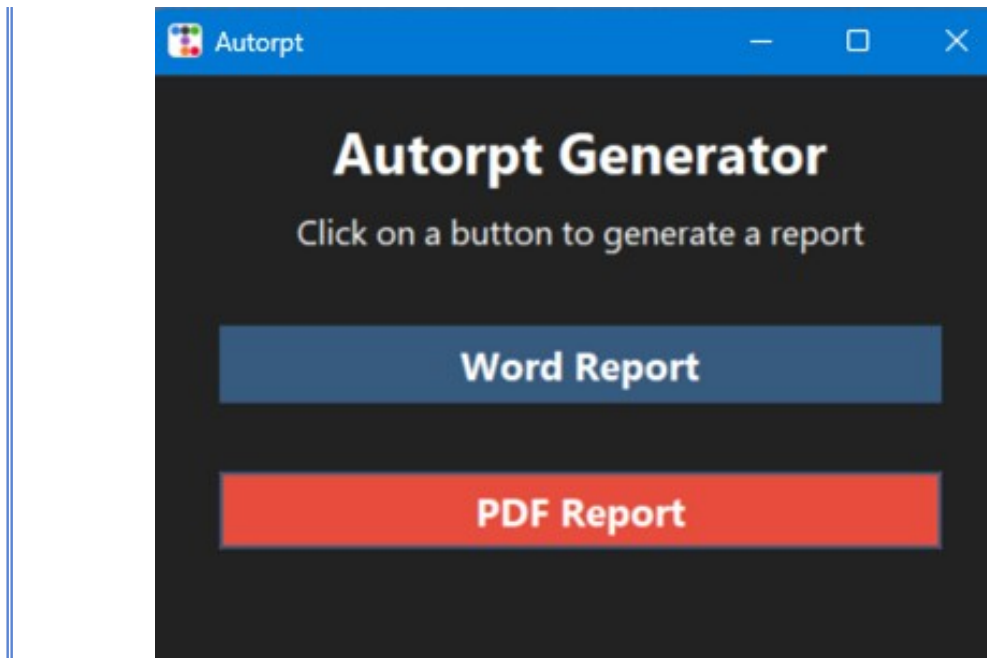


Figure 1: Graphical user interface for autorpt.

Select the report you want to create with content.md and budget.xlsx downloaded from the same [repository](#) that houses the gui.exe and stored in the same folder that you saved the gui.exe and run the executable.

Best Practices

- Use standardized templates provided by the funder—CAL FIRE provides a template for you to use as a payment reimbursement form.
- Keep records organized and audit-ready—always prepare your documents with the thought of an auditor in mind.
- Always communicate with your grant manager about any issues or changes.

DRAFT Startup Pathways

Vance Russell

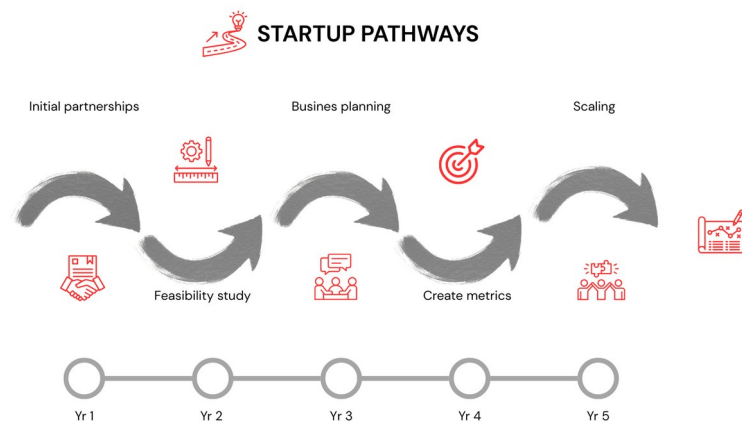


Figure 1: Sample startup pathways.

Takeaways

Following steps from concept to implementation with a checklist of to-dos at each stage is an ideal way to structure your startup pathway. Modify the steps, checklist, and waypoints to suit your startup's needs (Figure 1). Some key takeaways for the wood product businesses and organizational program/project startup include the following:

1. **Program design vs. startup.** Understanding the distinction between designing a program and launching a business is critical to aligning your development pathway with the right resources and expectations.
2. **Stress testing feasibility.** Rigorously testing your concept through scoping and feasibility analysis prevents costly mistakes and ensures your business model can withstand real-world conditions.
3. **Financial vulnerability.** Wood products businesses face unique financing challenges due to limited grant opportunities, long capital timelines, and the need to avoid inappropriate funding sources such as venture capital.
4. **Challenging process.** Navigating infrastructure gaps, securing long-term feedstock contracts, obtaining permits, and finding skilled leadership requires patience and strategic planning.
5. **Networking & collaboration.** Building partnerships to share resources, equipment, workforce, and infrastructure can provide mutual support and accelerate success in this sparsely connected industry.

Introduction

Program design and the feasibility analysis steps for a startup are connected processes, but they differ in important ways. Program design focuses on establishing vision, goals, and the conceptual framework for what you want to achieve, while feasibility analysis rigorously tests whether that vision can be practically and financially realized. For wood products businesses in California, this dual approach is essential given the unique challenges of securing feedstock contracts, navigating complex permitting processes, and securing appropriate financing. The startup pathway outlined in this chapter provides a structured progression from concept to implementation, with checkpoints to stress-test your assumptions at each stage. By matching the right resources, expertise, and capital to each phase of development, one can navigate the challenging landscape and build a sustainable forest-based business.

Wood Processing & Native American Tribes

Wood processing programs led by Native American Tribes are an important, and occasionally overlooked, area of wood products. Tribal-led wood processing programs are often multi-benefit operations that

intertwine directly with providing wood products for Tribal citizens and community members, workforce development for Tribal citizens and community members, and land management or restoration goals. While these operations vary in size, they face many of the same challenges as business or nonprofit wood-processing start-ups. Contact the Forest Business Alliance team for additional resources on Tribal-led wood processing programs.

Challenges

We often go straight to funding when it comes to startup challenges, but funding can be symptomatic of other issues that hinder forest businesses from getting up and running. Some of these include

1. **Lean funding & finance opportunities.** Many grant opportunities are geared towards nonprofits. CAL FIRE's Business and Workforce Development Grants and USFS's Wood Innovations Grants are some of the few that offer businesses the opportunity to apply for public funds.
2. **Lack of infrastructure.** Access to power, other mills, replacement parts for mills, maintenance workers, or other infrastructure critical to your business function can make or break how efficiently and profitably you run.
3. **Inability to facilitate long-term feedstock agreements.** Swezy (2025) mentions that once the sawlog supply normalized following the Dixie Fire, that although the regional Forest Service leadership is trying to align with the Wildfire and Forest Resilience Task Force Action [Plan](#) for treatment targets, these objectives do not often match the local district ranger capacity to secure feedstock contracts nor the funding for the work resulting in inconsistent log supplies and financial instability for wood processing facilities.
4. **Sparse networking and partner opportunities.** This is important for nonprofits in particular, as well as for businesses working with collaboratives. At another level, partnering to share resources, investment, equipment, workforce, and infrastructure can mutually support new businesses.
5. **Workforce to run wood processing facilities.** Although experienced labor to run mills may be available, finding skilled leadership for supervisory positions to support operations and grow startups may be difficult and has negatively affected the long-term viability of new mills and processing facilities.

Funding & Finance

A critical component of funding and finance is matching the business development stage Table 1. Note these are broader stages than the startup pathways stages and therefore have slightly different names. Aligning resources and advisory roles in this fashion can help accelerate development through the right kind of funding or finance.

VCs/Equity Finance $\neq$ Wood Products Businesses

- **Venture capital** is almost never appropriate for wood products due to the high return demanded vs. the low return for wood products businesses. VCs are not usually interested in pre-revenue startups.
 - **Equity financing** is also not typically appropriate for small businesses typical of the forest industry. As a small business, you would have to pay the financier profits annually or, over time, buy them out. On the plus side, with equity financing, you don't have a loan repayment schedule.

Table 1: Match your funding/finance to your business stage (may be different from the pathways stages). Pilot: years 1-2; Early Revenue: 2-4. Adapted from the Do Good CFO slide deck from the Startup Pathways Workshop. Note these are broader stages than the startup pathways stages.

Grant Funding

Grant funding is an excellent source of funding to start businesses and test any pilots you have in mind. Please see the proposal and planning chapters for more details, and consult various online grant request proposal portals for available grants. A golden rule of funding is creating a singular strategy based on three layers: 1. grants/incentives; 2. debt; and 3. equity.¹ Maximize grants first, then take on only the debt you can service, considering equity last or not at all (Figure 2).



Figure 2: Three layers, one strategy for funding startups. From Do Good CFO.

Debt Options

Table 2 shows the various debt finance options available and appropriate to forest-based businesses.

Table 2: Summary of multiple debt options. Adapted from Do Good CFO.

¹Neal Gottlieb, personal communication, 2026, Startup Pathways Workshop.

Financial Due Diligence

When considering any funding or financing options, always review the fine print in contracts, have your lawyer review them, and do your due diligence for investors or equity investors, e.g., get to know who you're "marrying".

Deal structures

Examples of blended finance deal structures are shown in Table 3.

Table 3: Illustrative deal structures. Adapted from Do Good CFO.

Startup pathways

We provide a walk through six stages for developing a wood products startup, from brainstorming your initial concept through implementation and scaling. Each stage includes specific checkpoints to test assumptions and prepare for the next phase. Adapt this progression to your business's unique constraints and opportunities.

I. Concept

The concept stage is brainstorming or coming up with the idea that solves a key problem. It is the spitballing stage for the initial ideas that precedes the pre-feasibility stage. Once you get this down on your napkin, it's critical to stress-test it and share it with key people and users to see if it will actually be used.

II. Scoping

In addition to assessing whether your idea will work, scoping is a pre-feasibility phase that should cover a range of items, from a basic project or program outline to financials and permitting. Some items to consider include the following:

- Establishing a vision and project concept, goals, and objectives
- Identification of technology options and the preferred option
- Site and infrastructure evaluation
- Feedstock volume and pricing assessment
- Identification of the permitting pathway

- High-level financial analysis (capex, opex, potential sources of capital, market demand/pricing, sensitivities)
- Partner mapping/initial engagement
- Establishing a preliminary project schedule

Pre-feasibility is really about testing the concept, and if you can check these boxes, it's time to move to feasibility and deepen your analysis.

III. Feasibility

Some businesses hire consultants to help with feasibility studies, financial analysis, market analysis, and securing feedstock supplies or permits. Some other items to consider include

- Securing site control
- Preliminary engineering and design for the preferred tech, and a third-party review of it
- Detailed financial analysis
- Establishing the appropriate business structure
- Further feedstock and market analyses, if needed
- Securing feedstock supply letters of intent/contracts
- Securing offtake letters of intent/contracts
- Identification and initial pursuit of capital stack
- Refine and maintain the Gantt
- Partner/stakeholder engagement
- Develop pitch deck, white paper, and web landing page

Continuing to deepen the project or business financial analysis is critical at this stage.

Another critical component is securing a site and working with local agencies, the county, and other compliance authorities to secure permitting needs. Permits can torpedo any project, and having them in hand, or at least well down the path to completion, is critical here.

IV. Late-stage

Steps in this stage are the final touches to put your project planning and feasibility analyses in place and to prepare the organization for implementation. Some final steps include

- Final engineering/design
- Construction and operating permits
- Complete lender/investor requirements and financial close
- Establish construction schedule and plan
- Procure long-lead equipment
- Hone pitch deck

V. Implementation

Time to take the plunge! Implementation is busy and can completely envelop a business, so don't forget to pause, reflect, and go back to the above. Make sure you're implementing as you planned. Check in with your funders and get feedback, but don't let them micromanage the project.

VI. Stock Taking & Scaling

We recommend stock-taking and results analysis at each stage of the startup pathway. It's also a good idea to analyze outcomes annually, determine what was learned from them, and make adjustments. Communicating to your key funders and audiences about this process and your learning is an open and honest way to gain support and feedback on the work and process underway.

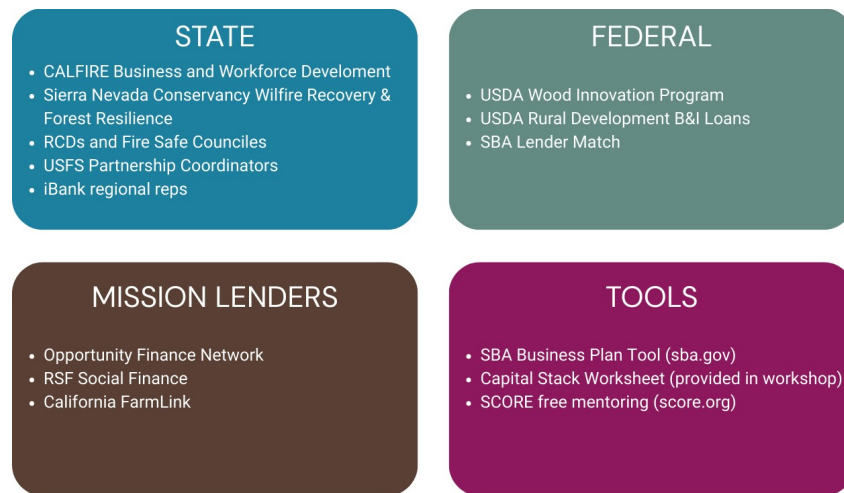


Figure 3: Key resources for your capital journey. Adapted from Do Good CFO Startup Pathways slides.

At the same time, start considering how to scale your business. A more detailed look at scaling can be found in Chapter 9.

Resources

- **Key Resources.** A summary of key resources from the Startup Pathways workshop is shown in Figure 3.
- **Forest Business Alliance Startup Pathways Workshops.** FBA held a series of workshops in March 2026 focused on mobile modular wood processing, startup pathways, and financing options. This chapter summarizes the key points from those workshops. The finance portion of the workshop is available as a recording on the FBA YouTube Channel. Please see the [workshops](#) page of the FBA website for links to the recording and agendas.
- **Wood Products Campus Siting and Financial Risk Analysis.** As part of the South Central Sierra feedstock aggregation pilot project (<https://californiaforestproducts.org/fapps/>) sponsored by the California Governor's Office of Land Use and Climate Innovation (LCI), Wildephor Consulting Services has published a siting and financial risk analysis [report](#). The study included three elements: 1) a biomass feedstock supply analysis; 2) a biomass feedstock storage model; and 3) a biomass project financial risk analysis to produce the Biomass Utilization Financial Feasibility and Sensitivity (BUFFSTM) analysis.
- **Feedstock Pilots.** The Governor's office of Land Use and Climate Innovation funded 5 [feedstock aggregation pilots](#) to accelerate the long-term availability of wood feedstock for processing into a variety of products. Several of the pilots formed [Joint Powers Authorities](#) to create local entities that help enable long-term feedstock contracts. The formation of these entities is instructive for startups, and links to them are useful to sustainably source wood.

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DRAFT Grant Closeout

Vance Russell, Christopher Sacco, Regine Miller, Kelly Kahling

Takeaways

1. Grant closeout is more than just a report; it offers an opportunity for internal learning and external outreach for your business or organization.
2. Closeout includes complete key processes required by your grant agreement and institution. Make sure to start grant closeout well ahead of the end of your grant.
3. For CALFIRE grants, closeout involves both financial and project report as well as completing forms sent to you by CALFIRE and on eCivis.

Process

It is important to ask two key questions before starting closeout:

1. Do you need to amend your contract, based on task completion, expenses, or other factors? If yes, see the Amendments callout. If not, proceed to consider the next question.
2. Are you ready to close out the project? Have you completed the deliverables, spent whatever grant funds you plan to spend, and fulfilled all the contract terms? If yes, read on. **Remember:** You will not be able to submit further activity reports or complete pending tasks after the grant is closed and any unspent funds will be de-obligated and returned to the grantor.

Amendments

Amendment procedures differ across funders and sometimes across grant programs within the same agency, yet they almost always take longer than expected. Plan for any adjustments you may need well in advance of your grant closeout date. Additional tips to keep in mind include the following:

1. **Grant manager.** Always consult with your grant manager before starting an amendment.
2. **Submittal.** For CALFIRE, submit amendments 90 days from the agreement expiration date. For other programs, see the terms in your contract.
3. **Timing.** Amendments take time to process. To be safe, you should allow at least 6 weeks from submittal to approval.
4. **Invoicing.** Once an amendment is submitted, eCivis cannot accept invoices until the amendment is approved.
5. **Cash flow.** To avoid cash flow problems, try to time submittal immediately after submitting an invoice.

Grant closeout is a process initiated well before your agreement termination date (Figure 1). The hypothetical timeline of approximately three months to complete closeout applies to CALFIRE and other grants, dependent on grant terms.



Figure 1: Hypothetical grant closeout process with timeline started three months prior to the grant end.

Once you have determined whether or not you will need to request an amendment, and submitted one if you do, start gathering final project metrics & outcomes. Three months ahead of time may seem early to start, but you may need to track down data, analyze it, and start thinking of ways to visualize it in a compelling manner.

Two months before the grant report is due, draft the report with the project team. Note gaps in information, identify key learning, and things you might change, and treat the report writing as an overall process of learning within your organization. At this time, identify pathways to leverage your grant with other resources (you may actually want to start this process at least a year before the grant ends) and opportunities to use the grant report results to communicate with other audiences.

For example, you may want to tie results from the project into your annual report or a blog post on LinkedIn describing what you learned from the funding and the project. Use this as an opportunity to consider ways to thank your funder(s) and to interact with them to secure further funding (if needed).

One month prior to the grant's end, analyze and process financial information. Don't leave this until the last minute, as it takes time, and you don't want to be combing through a mountain of receipts and other expenses at the last minute. You can always add expenses as you close out, but doing fewer at a time is easier.

Submit the final report and celebrate your team's success. Note that grant reports may be due sometime following the end of your contract term, but it's best to submit them as close to possible at this date, given other project priorities, project spending, and other distractions that may cause reports to be submitted later than expected.

Best Practices

Broadly speaking, closeout has a number of best practices that include:

1. **Get organized.** Review grant commitments (including project goals), approved budget, and expenditures to date. Create a timeline for finishing the report that includes cut-off dates for implementation/expenses and staff responsible for each item. Gather all relevant documentation on project accomplishments, metrics, photos, and financials.
2. **Analyze.** What did you get right, what methods worked/didn't work, what did you overestimate? What was the project's impact on the ground and for your organization?
3. **Be transparent.** Be constructive and honest in your reporting on what worked and what you would do differently. Think of the take-home results as pluses (what went well) and deltas (what would you change next time).
4. **Visualize.** Think of creative ways to visualize and convey your results beyond words in a paragraph (within bounds of agreement and reporting requirements). Graphs, maps, and photos all help communicate your message and the accomplishments of the project.
5. **Learn.** Spend time reviewing and learning from results. Apply learnings to ongoing projects and measurements to the overall business/organizational plan.
6. **Share.** Share your results beyond the report submitted to CALFIRE. Also, the work you do to compile the report can serve multiple purposes, e.g., communications or an annual report.

Attention

If your project results, reports, or analyses contain proprietary information, discuss with your program manager how to submit and share results that do not compromise or put this information in the public sphere. Some businesses have not submitted results, but still showed proof of completion, e.g., feasibility or marketing studies.

CALFIRE Closeout

In addition to financial and programmatic reporting, you will need to complete additional materials:

1. Complete the Project Closeout Report Form provided to you by CALFIRE. Be sure to include allowable supporting materials, such as photographs, news articles and reports, that demonstrate your project's accomplishments.
2. Complete the eCivis portal closeout process. This form in eCivis also serves as your final Financial Report (Figure 2).
3. Make sure to carefully organize and create backup documentation for the project. Besides having it for your files, once you submit the closeout to eCivis, your project information is no longer accessible.

The screenshot shows the 'Award Dashboard' in the eCivis portal. At the top, there are several buttons: 'View Budget', 'View Goals', 'View Files', 'Submit Financial Report', 'Submit Activity Report', 'Request Grant Amendment', and 'Award Closeout'. The 'Award Closeout' button is highlighted with a red circle and a red arrow pointing to it. Below the buttons, there is a section for 'Award Detail' containing various fields: 'Subrecipient', 'Program: Business and Workforce Development 2022 Q1/Q2', 'Award ID', 'Award Status: Awarded', 'Approval Date: 02/02/2023', 'Obligation amount', 'Award amount', 'Total Non-Federal Award', 'Total Match: \$0.00', and 'Performance period: 12/09/2022 - 03/31/2026'.

Figure 2: Award dashboard for eCivis showing the award closeout button to start the closeout process.

Resources

1. Examples of forms and documentation for closeout can be found in the Grant Closeout Workshop on the [FBA youtube channel](#).
2. Contact [forestbusinessalliance @ gmail dot com](mailto:forestbusinessalliance@gmail.com) if you have any specific questions on closeout, eCivis, or reporting.

Forest Business Alliance Takeaways

Table 1: Summary of pluses (what went well) and deltas (what we would change) for FBA Phase I. Current pluses/deltas listed are for illustrative purposes only and will be updated following a more detailed review.

[illegible]

Created with Curvenote

Metrics

Table 2: Summary of key metrics and accomplishments for FBA Phase I. CALFIRE Business and Workforce Development Metrics, along with FBA-developed technical assistance and capacity-oriented metrics tracked during FBA Phase I. Percent achieved amounts are from proposed targets.

[illegible]

Created with Curvenote

We conducted a review of the Forest Business Alliance through internal and external processes to assess program effectiveness and inform future directions. External review consisted of personal communications and surveys distributed to FBA participants to gather feedback on their experiences and outcomes. Internal review took the form of a facilitated staff meeting that systematically examined the pluses and deltas of each program component and carefully considered next steps for program development.

Startup Progression

Throughout Phase I of FBA, we tracked stages of contacts from initial identification to funded proposal status (Figure 1). Other stages for collaborators, such as source and participation, were included to account for other types of network participation. Identified, participating, and source category numbers were high at the beginning of the project, and technical assistance was steady but increased dramatically in 2025 as more applicants became aware of the Alliance and contacted us for assistance.

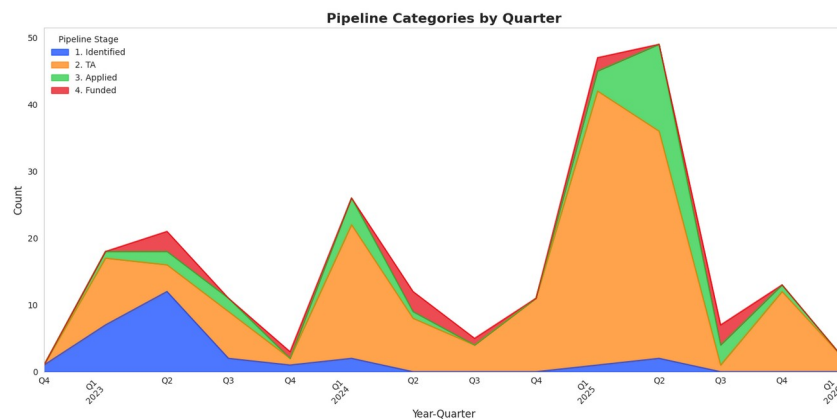


Figure 1: Progression of FBA pipeline technical assistance.

Examined in a different manner, cumulative technical assistance rises dramatically over time, while numbers for other pipeline stages roughly stay the same Figure 2.

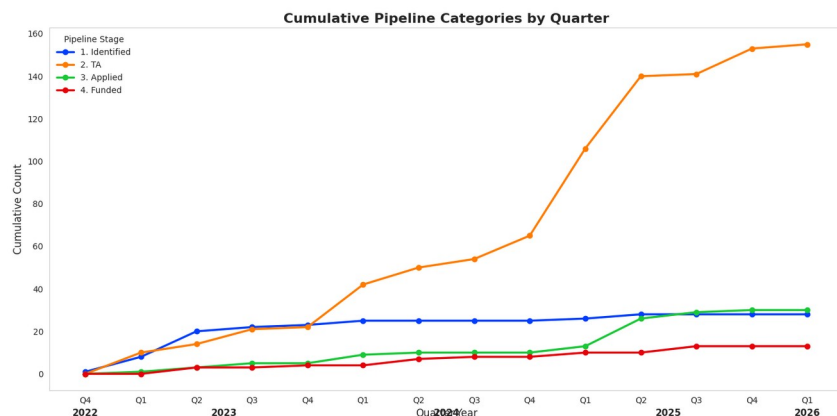


Figure 2: Cumulative pipeline categories over time.

Technical Assistance

Technical assistance proved to be a core strength of FBA Phase I, with direct support to BWD applicants, specialized Tribal TA, and comprehensive resources, including funding and finance options, and geospatial/contact databases for wood products businesses, helping to build sector capacity. However, several opportunities for improvement emerged: TA delivery needs to be more strategically targeted, a funding gap was identified, office hours saw sparse attendance, and expansion into additional statewide programs is needed. These deltas suggest that while the foundation of TA provision is solid, the program should refine its delivery mechanisms and extend its reach to maximize engagement and impact.

Workshops

FBA delivered 39 workshops to 559 participants, exceeding the target by 195% and total participants by 112%, with tribal workshops and funding & finance sessions drawing the strongest attendance. Participation varied significantly across workshop types and timing, reflecting inconsistent interest or scheduling challenges. Workshops tied to grant deadlines consistently saw the best turnout, suggesting participants prioritized sessions when facing immediate funding deadlines. Funding & finance workshops resonated most because applicants needed specific knowledge to complete proposals for entities with money. Phase II should anchor the workshop schedule to grant cycles and continue emphasizing tribal and funding-focused offerings.

Peer Learning Network

"I left each Peer Learning Network meeting feeling inspired and made contact with many members afterwards." Mika Cook, Sustainable Rural Forest Community Strategist, Humboldt

Good participation around CalFire grant deadlines—this probably indicates that PLN was perceived as a strong CalFire support group specializing in grant funding. The in-person PLN meeting at Sierra Institute was well attended, and participants developed strong peer-to-peer and organizational connections (though qualitative, hard to document). Funding for in-person field trips is limited. Change to business-specific network, e.g., mills or biomass only sessions.

"The amount of experience in the wood products industry to lend ad hoc expertise is limited. I really appreciated the speakers, fieldtrips, and networking provided by the Network." Temra Costa, Regenerative Forest Solutions

Next Steps

It is clear there is a need for capacity building and technical assistance for businesses and nonprofits in the wood products sector. In particular, needs are highest around organizational structure/mission, business planning, proposal writing, funding, finance, and grant administration. The following steps are based on feedback from businesses and nonprofits that FBA is assisting, as well as our collective experience over the initial project term. The following are the highest priority items:

1. **Focus on technical assistance & key workshops.** Direct technical assistance was one of the most effective forms of support FBA provided. The challenge was connecting with new and existing organizations, but this has greatly improved over time. Workshops were best attended during grant rounds and when they focused on proposal development or funding, especially funding/finance from entities with money (agencies, banks, foundations, private investors).
2. **Reconfigure the Peer Learning Network or focus on specific topics with similar businesses.** Tribal PLN was well attended, key reasons why? PLN was well attended at grant rounds or with key speakers (e.g., River City Bank). But attendance was low in the latter half of Phase I. Focusing on a network of businesses that need it, or creating ad hoc networks when the need arises, might be a better approach.
3. **Continue funding Tribal Workshops.** These have been very well attended and in high demand for a variety of topics. Creating additional or follow-up resources with the workshops could supplement capacity building.
4. **Increase connection to unsuccessful grant applicants.** Last year, FBA reached out to more previously unsuccessful grant applicants and helped several secure funding through key adjustments to their program design and grant-writing approaches.

5. **Continue communications & outreach.** Each year, more people know about FBA and reach out, either unsolicited or at the advice of CALFIRE staff, for technical assistance. Having outreach materials such as the guide, website, and webinar/workshop recordings is not just an incredible learning resource but also a library for providing assistance.

Additional review of the difference in Peer Learning Network participation between the general PLN and the Tribal PLN would be helpful. Systematic outreach to new startups and nonprofit programs is a feature of the technical assistance that should continue, especially in regions with fewer applicants or lower capacity to apply.

Reviving Sonoma's Timbershed

Christopher Sacco

Takeaways

- **Think Holistically:** Make sure you understand how raw material moves from forest to finished product and seek to design a closed-loop system. Integrating forest restoration, processing, and product design into one continuous flow ensures quality control, reduces waste, and creates reliable supply chains.
- **Leverage Partnerships:** Collaboration with universities, nonprofits, and public agencies can accelerate innovation and help overcome technical and financial barriers. Research partnerships validate performance, while grants and local alliances provide credibility and resources.
- **Demonstrate Through Pilots:** Visible demonstration projects—whether prototypes, small structures, or public installations—help generate awareness among potential future customers and build market confidence. Prioritize pilots that showcase feasibility and invite stakeholder engagement.
- **Anticipate Policy Barriers:** Regulatory frameworks often assume and favor industrial-scale operations. Restoration-driven enterprises must plan for permitting hurdles and advocate for tailored pathways that recognize ecological and social benefits.
- **Promote Stewardship through Prosperity:** Linking employment and other economic benefits directly to forest restoration fosters local buy-in and helps ensure that ecological restoration is understood not as a burden, but as a shared opportunity for prosperity and long-term sustainability.

Background

A century of unsustainable logging followed by decades of aggressive fire suppression have left much of California's forestland dangerously overgrown and lacking diversity — conditions that fuel catastrophic wildfires, threaten communities, and erode the long-term resilience of the state's forest ecosystems. In Sonoma County, these vulnerabilities have been tragically underscored by multiple wildfires, including the Tubbs (2017), Kincade (2019), and Glass Fires (2020), which together destroyed thousands of homes, displaced tens of thousands of residents, and caused billions of dollars in damages—leaving lasting scars on the County's people, economy, and forests.

Forest management practices—such as thinning overcrowded stands, conducting prescribed burns, and reducing fuel loads—are critical to restoring forests into more resilient, low-density structures less prone to wildfires. Yet in Sonoma County and across California, scaling these practices remains difficult. Current market incentives do not support the removal of small-diameter or non-merchantable timber, as the costs of harvesting, hauling, and processing typically exceed potential returns. In addition, sawmill processing facilities are limited, creating a general bottleneck for wood processing across size classes and species. Without new value streams for low-value wood and processing capacity, large-scale fuel reduction and ecological restoration will remain financially out of reach, constrained by the limited public funding that can support them.



Figure 1: Jeremy Fisher, Founder/CEO, Forestree Collective at the Forestree wood shop.

Reviving the mill

Forestree Collective's mill in Sebastopol, CA, and woodshop in Penngrove, CA, create a circular, market-based approach to forest restoration (Figure 1). The 172-acre historic mill property has been upgraded to include a Woodlandia 200XP single-pass log ripper to process small-diameter logs, a Woodmizer LT-40 Wide portable bandsaw mill for larger logs, and a compact telehandler. Forestree also has a fully equipped 5,000-square-foot woodshop for secondary wood products manufacturing, including a kiln drying operation with a solar kiln built from their own mass timber product, nail-laminated timber (NLT), and a dehumidification kiln. Shop equipment includes table saws, jointers, planers, panel saw, bandsaw, shaper, and a wide belt sander.



Figure 2: A log harvested during restoration at Forestree Collective's mill in Sebastopol, California.

Forestree's vertically integrated process functions as a closed-loop system. Small-diameter trees (6 inches and larger)—responsibly sourced through partnerships with local landowners and foresters—are milled in-house,

solar-kiln-dried, and transformed into value-added products in the woodshop (Figure 2). Each stage of production builds on the one before it: restoration yields raw material; milling produces dimensional lumber; the woodshop converts that lumber into finished goods. This integrated flow ensures consistent quality, reduces waste and energy use, and provides a reliable supply chain for climate-smart materials.

Restoration by design

Forestree Collective is redefining how restoration wood moves through the value chain by linking forest health directly to wood products innovation. Through a fully circular model, small-diameter and underutilized logs removed during thinning and fuel-reduction projects are transformed into durable, high-value products that extend the material's life, lock carbon into long-lived form, and create new incentives for stewardship (Figure 3). Forestree is currently advancing two complementary product lines—climate-smart furniture and custom interiors, and an innovative mass Nail-Laminated Timber (NLT) system, each expanding the circular model and opening new markets for regenerative, restoration-driven wood products.



Figure 3: Forestree's Sonoma Table is a modern take on the classic trestle table with the rounded corners for a contemporary look.

Climate-smart furniture and custom interiors are Forestree Collective's first expression of its circular, restoration-driven approach to material innovation. Restoration-grade material from fuel-reduction and forest health projects is turned into high-quality furniture and architectural elements for commercial and residential spaces.

Design and craft are integrated from the outset: Forestree's architects, designers, and fabricators apply species-specific milling and woodworking techniques to create durable, regionally distinctive pieces that embody place, lock carbon into long-lived form, and connect end users to the landscapes that produced them. Beyond the finished products themselves, this approach builds new economic pathways by converting low-value and underutilized biomass into premium goods that help fund restoration work, support local mills and craftspeople, and expand markets for diverse, regionally sourced species.

In parallel with its furniture and interiors line, Forestree Collective is developing an innovative **Nail-Laminated Timber (NLT)** building system that applies the same circular logic to mass-timber construction (Figure 4). NLT offers a low-barrier, adhesive-free mass-timber solution that is well-suited to small-scale and decentralized milling operations. Forestree's model enables restoration wood to be processed near where it is harvested in Sonoma County, reducing transportation costs, retaining more value in the local community, and creating viable markets for small-diameter and underutilized logs that are often burned or left to decay.



Figure 4: Forestree NLT panels on the ceiling.

Despite its inclusion in building codes and long history of use, NLT remains underutilized due to gaps in research, manufacturing guidance, and real-world demonstrations. Forestree is addressing these barriers through a partnership with the Berkeley Wood Lab at the University of California's College of Agriculture and Natural Resources (UC ANR) and the nonprofit Regenerative Forest Solutions. Together, the team is advancing structural research, optimizing low-cost fabrication methods, sustainable procurement, and producing demonstration projects that validate performance and expand market confidence. Their pilot NLT project was a solar kiln using Lignoloc wooden nails as fasteners (Figure 5).



Figure 5: Forestree's NLT solar kiln.

Current R&D focuses on using locally sourced, non-merchantable timber—milled and kiln-dried by Forestree—then assembled into prototype panels for testing. Dr. Paul Mayencourt of the Berkeley Wood Lab is working with Forestree Collective to help balance cost, ecological value, and carbon benefits—for example, by strategically intermixing species, grades, and dimensions to maximize use of lower-grade material while meeting performance requirements. Regenerative Forest Solutions conducted a feasibility study that Forestree is using to inform the procurement of material for its primary and secondary wood products.

Partnerships

Forestree Collective's collaboration with public agencies, academic institutions, nonprofit organizations, and local communities has been a major driver of its early success. The company's first success was winning the Sonoma County Bio Business competition in 2021, which helped launch the business, elevated its profile, and provided financial and technical resources.

The California Department of Forestry and Fire Protection (CAL FIRE) and the U.S. Forest Service (USFS) have recognized and supported Forestree's innovative work with grants that have helped Forestree expand its milling and manufacturing capacity and enabled it to pilot its NLT building system.

Collaboration with the UC Berkeley Wood Lab has proven vital to Forestree's advancement of its NLT system. From structural testing to design applications, this partnership bridges science, craft, and sustainability, helping to shape a more resilient and diversified wood economy and ensuring that research outcomes are translated into practical applications for the regional wood products sector.

Local government and nonprofit partnerships also play an important role in Forestree's work, including its stewardship of the 85-acre forest adjacent to the mill site. With CFIP funding and a forest management plan in place, Forestree collaborates with Fire Forward, the Gold Ridge Resource Conservation District, Regenerative Forest Solutions, CAL FIRE, UC Cooperative Extension, and the US Forest Service.

Impacts

Forestree Collective's integrated approach—reviving critical infrastructure, creating durable markets for restoration byproducts, innovating new sustainable products, and generating local jobs—illustrates how healthier forests, safer communities, and local prosperity can advance hand in hand.

A central impact of Forestree's work is the revival of wood-products infrastructure in Sonoma County (Figure 6). Forestree Collective and Regenerative Forest Solutions have recently launched Timbershed, a public-private partnership that intends to reopen a mothballed mill and transform it into a flagship wood products campus for Sonoma County, with the capacity to process up to six million board feet (MMbf) of timber each year.



Figure 6: Custom restaurant interior using Forestree crafted wood products.

Preliminary figures from a recent feasibility study indicate a high-production NLT facility would use at least one million board feet of small-diameter timber annually—equivalent to treating roughly 250 acres of forest, or nearly 10% of the estimated yearly fuel-reduction output from non-industrial private forests in Sonoma County. By sourcing timber directly from local thinning and fuel-reduction treatments, Forestree Collective is contributing to wildfire resilience in its own backyard.

To date, Forestree Collective has created 3 full-time and 6 part-time jobs in forestry, log hauling, milling, and manufacturing sectors. At full capacity, the planned Timbershed wood products campus would generate an additional 40 FTE direct and 77 indirect jobs, further strengthening Sonoma County's economy and tying the community to sustainable forest management. These positions represent more than employment—they embody a commitment to community resilience and sustainable livelihoods tied to the stewardship of California's forests.

Challenges

Restoration-focused wood enterprises like Forestree Collective operate in a landscape shaped by rising wildfire risk, growing demand for sustainable materials, and decades of underinvestment in rural forest-products infrastructure. These businesses must navigate fragmented supply chains, limited processing capacity, and high operational costs. At the same time, they face uncertain markets for small-diameter logs and by-products, which remain inconsistent or underdeveloped despite being the primary outputs of ecological thinning. When sourcing from small private forest landowners, added administrative requirements and permitting complexity can slow project timelines and increase overhead costs. Tight margins and limited access to capital further constrain their ability to invest in equipment, facilities, or workforce expansion.

Layered on top of these structural challenges are California's Forest Practice Rules (FPRs), which were designed to prevent destructive logging but apply uniformly to all operators regardless of scale or intent. For landowners with substantial timber resources to manage, this means navigating costly Timber Harvesting Plans that require professional foresters, legal review, and months of permitting — costs that exceed the value of the small-diameter timber involved. Approval timelines and permitting review processes are notorious for delaying urgent wildfire-recovery and fuel-reduction work, and led to a temporary pause in requirements in CA in 2025 to speed implementation that has generally lagged behind state-mandated goals. Because the rules were built around large commercial harvests, they offer limited flexibility for the commercialization of material derived from ecological thinning, as the rules apply even to low-value or "waste" wood removed for safety, wildfire prevention, or

post-disaster cleanup. The lack of distinction and inability to make products directly with this material without specific permits make restoration-wood business models financially precarious.

Taken together, these structural, market, and regulatory barriers create a challenging operating environment for enterprises seeking to turn low-value timber into climate-smart, regionally sourced products. Without more tailored pathways, streamlined permitting, and policies that recognize the distinct role of restoration-driven operations, California risks constraining the very businesses capable of accelerating wildfire resilience, revitalizing rural economies, and unlocking new value streams from materials that would otherwise go to waste.

Replicable model

Forestry Collective's work is reshaping forest stewardship and generating new economic opportunities in Sonoma County while demonstrating a practical model that other rural communities can replicate to address California's interconnected wildfire, forest health, and carbon challenges. Important lessons from the company's work to date include:

- **Think Holistically:** Make sure you understand how raw material moves from forest to finished product and seek to design a closed-loop system. Integrating forest restoration, processing, and product design into one continuous flow ensures quality control, reduces waste, and creates reliable supply chains.
- **Leverage Partnerships:** Collaboration with universities, nonprofits, and public agencies can accelerate innovation and help overcome technical and financial barriers. Research partnerships validate performance, while grants and local alliances provide credibility and resources.
- **Demonstrate Through Pilots:** Visible demonstration projects—whether prototypes, small structures, or public installations—help generate awareness among potential future customers and build market confidence. Prioritize pilots that showcase feasibility and invite stakeholder engagement.
- **Anticipate Policy Barriers:** Regulatory frameworks often assume and favor industrial-scale operations. Restoration-driven enterprises must plan for permitting hurdles and advocate for tailored pathways that recognize ecological and social benefits.
- **Promote Stewardship through Prosperity:** Linking employment and other economic benefits directly to forest restoration fosters local buy-in and helps ensure that ecological restoration is understood not as a burden, but as a shared opportunity for prosperity and long-term sustainability.

Forestry Collective's experience shows that restoration-based enterprises can do far more than process wood—they can realign local economies with ecological health, rebuild critical rural infrastructure, and demonstrate a viable path toward wildfire resilience and climate-smart development. The lessons emerging from their work underscore the importance of integrated systems, strong partnerships, visible proof-of-concept projects, and proactive engagement with policy barriers. Together, these insights point to a broader opportunity: with the right support and regulatory flexibility, communities across California can adopt similar models that turn restoration needs into durable economic and environmental benefits.

About

Forestry Collective, founded in 2021 by building scientist Jeremy Fisher and architect Marisha Farnsworth, is pioneering circular, market-based solutions to the wildfire crisis. This case study highlights Forestry's work to rebuild Sonoma County's wood products infrastructure and to transform non-merchantable timber and restoration byproducts into climate-smart, value-added materials. By creating new economic incentives for stewardship, Forestry contributes to reducing wildfire risk and restoring forests while opening pathways for rural economic renewal.

Credits

We commend the innovation and leadership of [Forestry Collective](#) for advancing a practical, circular model that transforms restoration-grade timber into high-value, climate-smart products while strengthening forest health, rural economies, and wildfire resilience. A special thanks to Regenerative Forest Solutions for contributing to this article. We also extend our deep gratitude to [CALFIRE](#) for funding the Forest Business Alliance's work with Forestry Collective, and other community forestry partners across California. Rural and Tribal communities engaged in the forest products sector can benefit from resources such as the [Forest Business Guidebook](#), the Forest Business Alliance [website](#), and [workshops](#), including the Early Stage Project Development workshop.

Photos courtesy of Forestry Collective.

Plumas Wood Fiber

Christopher Sacco

Takeaways

- **Addresses multiple problems.** Simultaneously addressing societal/environmental problems and industry needs makes for a compelling case to governments, grantmakers, and impact investors.
- **Importance of grants & partnerships.** Grants and partnerships are essential to offset business risk, especially when resources are constrained.
- **Connection to research.** Strong academic collaboration and rigorous lab testing are necessary to validate product performance.
- **Broad outreach.** Broad outreach and participatory validation help to generate interest and move potential buyers from curiosity to commitment.
- **Scaling.** Careful planning—such as partnering with energy providers and securing early buyer commitments—helps reduce financial risk and stabilize cash flow when scaling from pilot to commercial production.

Context

For decades, western U.S. forests have accumulated undergrowth and small-diameter trees because of fire suppression and past management practices. That accumulation increases wildfire intensity, killing trees and releasing large pools of stored carbon. At a larger scale, this is contributing to ecosystem destruction. Removing that excess biomass is expensive and often produces large volumes of chips with little commercial value.

At the same time, the horticulture industry relies on vast quantities of potting substrate, historically supplied by peat moss harvested from peatlands, which destroys peatland ecosystems and their carbon stores. Peat moss harvest also releases large amounts of greenhouse gases, degrades biodiversity, and causes long-term habitat loss. Additionally, logistical challenges and climate impacts have disrupted summer peat harvests in Canada—for example, yields fell by 25–50% in 2023—producing wide swings in peat price and availability; this volatility is a major concern for the horticulture industry, which expects to double or quadruple its use of potting substrates in the coming decades.

Plumas Wood Fiber emerged at the intersection of these trends. The company sees market demand from California's growers and substrate blenders for a locally sourced substitute that matches peat's horticultural performance while avoiding peat's environmental and supply-chain drawbacks. Early analytical work with academic partners and funding from the Conservation X Labs Fire Grand Challenge pilot allowed Plumas Wood Fiber to explore whether western conifer species can meet the physical properties required for potting mixes. The preliminary results and stakeholder interest signaled a viable market and a compelling circular solution: chip material from hazardous fuels-reduction projects converted into a valuable horticultural input that reduces transport emissions and helps finance forest treatments (Figure 1).



Figure 1: Sample wood fiber substrate produced by Plumas Wood Fiber.

Business model

Plumas Wood Fiber's business model ties feedstock procurement directly to forest restoration activity. The company will source wood chips from contractors performing fuels-reduction and restoration work on federal and private lands, and purchase chips at prevailing market rates to ensure a steady supply.

The company's value proposition rests on three connected advantages:

1. **Feedstock.** Ready access to low-cost feedstock from wildfire fuels-reduction projects;
2. **Transportation.** Much shorter transport distances to West Coast horticultural buyers compared with imported peat, and
3. **Performance.** Validated substrate performance for a range of food crops and ornamental plants.

These factors combine to lower delivered cost and greenhouse-gas footprint while meeting the technical requirements buyers expect.

Plumas Wood Fiber is moving deliberately from pilot validation toward commercial scale. The pilot phase used grant funding to produce hammer-milled fiber in partnership with Real Good Soil and distribute samples to roughly twenty horticultural stakeholders and academic partners for growth trials. Those trials, together with lab testing led by a university horticultural substrates lab, form the evidence base required to convince large buyers to adopt a new substrate.



Figure 2: Plumas Wood Fiber using a hammer mill to produce sample substrate for growth trials.

In the pilot phase, Plumas Wood Fiber used hammer-milling (Figure 2) to produce sample fiber for growth trials (Figure 3). Operational scaling requires a purpose-built site and specialized equipment, including twin-screw extruders and automated baling lines that convert chips into an engineered fiber that mimics the drainage, aeration, and moisture-retention characteristics of peat (Figure 4).



Figure 3: Growth trials at Texas A&M University. Credit: Paul Winski.

In addition to extruders, production lines, and robotic baling and palletizing, commercial production requires a large chip-receiving yard and material-handling equipment. To reduce capital and operational expenses, the company is planning to co-locate its production facility with a biomass energy generation plant to be installed in Quincy, California, by West Biofuels of Woodland, California.



Figure 4: Fiber extruder at a wood fiber substrate production facility in Amsterdam. Plumas Wood Fiber's facility will include a similar production line.

Once large-scale production begins, Plumas Wood Fiber will target commercial nurseries and potting-mix blenders that buy substrate by the truckload, and it will also offer palletized bales for broader distribution to regional blenders and retail mix producers. The company projects a Freight on Board sale price near \$20 per cubic yard, producing delivered costs competitive with out-of-state wood-fiber suppliers once transport is included.

Startup

Plumas Wood Fiber's pathway to success is not without risk. Operationally, feedstock variability—moisture, particle size, seasonal availability—and the high capital intensity of extruder machinery raise the risk of production interruptions and cost overruns. Market adoption remains uncertain and will be shaped by the outcomes of growth trials, which are critical to overcoming conservative buyer hesitancy. At the same time, competitive pressure from established wood-fiber suppliers and lower-cost peat/coir blends could depress pricing and slow uptake, underscoring the importance of trial success in mitigating market risk. Regulatory and policy risk exists in multiple forms: changes in fuels program funding could shrink chip volumes; permitting for a processing facility can introduce delays or unexpected costs; and potential shifts in environmental policy—positive or negative—could alter incentives for peat displacement or biomass utilization. Finally, the business depends on a blended capital stack of grants, concessional loans, and commercial finance to bridge capex, so reduced grant availability or loan delays could postpone commercialization.

Nevertheless, if the validation work and early commercial runs confirm product performance at scale, Plumas Wood Fiber has clear expansion pathways. The model is replicable in other western forest regions with similar fuel treatment byproducts, enabling a distributed network of mid-sized facilities that feed regional horticultural markets. Product innovation could lead to proprietary blends tailored to specific crops, bagged retail products for consumer markets, or preconditioned fibers designed for large greenhouse operations. Technological improvements—especially gains in extruder energy efficiency and automation in baling and loading—would materially improve margins. Strategic partnerships with large offtakers, soil-blend manufacturers, and impact investors seeking wildfire- and carbon-positive projects could secure long-term revenue streams and flexible capital. Support in the form of grants or incentives that internalize the costs of forest restoration and carbon co-benefits could accelerate growth.

The horticultural industry is projected to significantly expand its use of substrates in the coming decades, with overall demand expected to double or even quadruple. Within that growth, wood fiber substrates are anticipated to see a tenfold increase in adoption—marking this as a high-potential sector where innovative start-ups are well-positioned to lead. However, establishing a viable industry for producing wood fiber substrate from Western forest biomass represents a novel approach for both forestry and horticulture. As a result, both sectors are currently adopting a cautious, wait-and-see stance.

Biomass enterprises inherently operate on tight margins due to the nature of the material—high volume, low value—and the logistical complexity of sourcing it. Building a viable business under these conditions demands careful planning, operational efficiency, sound partnerships, and strategic coordination. If Plumas Wood Fiber proves successful, it could unlock new opportunities for climate-smart forest management and sustainable horticultural supply chains. If not, it may dampen investor confidence and constrain future efforts to develop alternative biomass-based products. The stakes are high—but so is the opportunity.

Partnerships

Partnerships are central to Plumas Wood Fiber's strategy, providing technical validation, feedstock access, pilot capacity, and financing pathways that together de-risk scaling and accelerate market entry. On the research front, the company partnered with Dr. Brian Jackson at the Horticultural Substrates Laboratory, NC State Raleigh, to analyze western species—most notably Ponderosa pine—and confirm their compatibility and performance as engineered potting substrate (Figure 5). That scientific validation underpins the company's credibility with conservative horticultural buyers.

Figure 5: North Carolina State students working with wood chip samples from Plumas Wood Fiber.

Public and philanthropic support funded early study and piloting activities: the California Department of Forestry and Fire Protection (CAL FIRE), the Plumas County Fire Safe Council, and the USDA Forest Service provided grants for a feasibility study and pre-construction planning. Earlier this year, a Conservation X Labs Fire Grand Challenge award enabled a pilot collaboration between Plumas Wood Fiber and the Real Good Soil, which included procurement of a hammer mill to process Ponderosa pine chips into trial fiber that has recently been distributed to some 20 nurseries, academic institutions, and other horticulture industry stakeholders for growth studies.

Additionally, Real Good Soil, founded by Joel White, partnered with Plumas Wood Fiber to provide operational support during the pilot, offering feedstock sourcing, heavy-equipment handling, and onsite space that removed critical logistical barriers to testing. Real Good Soil's contribution accelerated sample production and distribution to industry partners while illustrating how local operational capacity can de-risk early-stage biomass processing.

Going forward, operational partnerships will be critical to Plumas Wood Fiber's success: supply contracts and working relationships with forest contractors and public fuels programs will be necessary to secure steady volumes of chips and help manage seasonality, while early offtake discussions and letters of support from nurseries and soil-blend companies identify clear commercial pathways. Similarly, strong financial partnerships—grants, impact finance, and public loan programs—will be key to assembling the capital stack needed for commercial operations. These collaborative relationships—scientific, operational, and financial—form the backbone of Plumas Wood Fiber's approach to turning excess forest biomass into a market-ready, peat-free substrate.

Impacts

Due to its environmental and community approach, Plumas Wood Fiber's project will have multiple positive environmental and social impacts both environmental and social. These are explored in the next two sections.

Environmental

Plumas Wood Fiber’s products provide multiple environmental benefits, beginning with their potential to reduce reliance on peat—a material whose extraction damages sensitive peatland ecosystems. Canada currently extracts and exports approximately 1.3 million tons of peat each year. Once removed from its native bog environment, where it serves as a long-term carbon sink, peat gradually biodegrades and releases carbon into the atmosphere that would otherwise remain sequestered. While wood fiber substrates also biodegrade, they are derived from renewable biomass already circulating within the active carbon cycle rather than sourced from a permanent carbon reservoir, making them a more climate-aligned alternative for horticultural and industrial use. Moreover, producing wood fiber substrate regionally avoids the emissions associated with long-distance peat imports, further reducing the product’s carbon footprint.

Plumas Wood Fiber’s model repurposes excess biomass from preventative fuel reduction and post-fire salvaging projects—material that might otherwise be burned, left to decay, or used in low-value energy applications—into a soil-enhancing product that temporarily stores carbon and improves plant-growing media. Commercial-scale substrate production supports wildfire mitigation and recovery by creating consistent market demand for biomass generated through fuels treatments and salvage harvesting. This demand can help finance additional forest management activities and incentivize contractor capacity to scale up forest-thinning and salvage operations. Under the planned facility configuration (two extruders operating on double shifts), Plumas Wood Fiber estimates it will utilize approximately 10,000 bone dry tons (BDT) of chips annually. While carbon accounting depends on variables such as avoided combustion, decomposition rates, and soil carbon dynamics, this volume represents a meaningful contribution to carbon management at the facility level.



Figure 6: Wood fiber storage bunker in the Netherlands.

Social

Establishing a commercial facility will generate local employment in production, logistics, and administration, while advancing economic development strategies in Plumas County—particularly those aligned with post-wildfire recovery efforts (Figure 6). Plumas Wood Fiber estimates that a facility operating under a double-shift configuration would employ six full-time equivalent staff to manage mill operations, bagging systems, marketing and administrative functions, coordination of wood chip deliveries, and outbound shipping of finished products.

Beyond direct employment, the company anticipates that scalable operations will support more stable, year-round job opportunities within the rural economy. Additional workforce demand is expected among forest-based contractors engaged in biomass removal, as the facility creates new market incentives to chip and transport

material. By providing a reliable destination for excess biomass, the facility transforms a disposal challenge into a catalyst for economic activity—linking forest restoration to job creation and regional resilience.

Lessons

Plumas Wood Fiber demonstrates how an integrated, regionally focused approach can turn a forest-management liability into an economic and environmental asset. Important lessons from the company's pilot phase include:

- Simultaneously addressing societal/environmental problems and industry needs makes for a compelling case to governments, grantmakers, and impact investors.
- Grants and partnerships are essential to offset business risk, especially when resources are constrained.
- Strong academic collaboration and rigorous lab testing are necessary to validate product performance.
- Broad outreach and participatory validation help to generate interest and move potential buyers from curiosity to commitment.
- Bridging the capital and operational gap between pilot-scale and commercial-scale production requires strategic planning, e.g., co-locating with energy partners and securing early letters of intent from offtake partners to reduce investment risk and smooth early cash flow.

Innovation

Plumas Wood Fiber has proven the technical and commercial potential of converting excess forest biomass into a market-ready, peat-free substrate, with partnerships and pilot results providing a credible path to scale. Achieving commercial success will depend on securing contractable chip volumes, reliable low-cost energy through strategic co-location, continued trial replication to win conservative buyers, and timely access to blended capital. With those elements aligned, Plumas Wood Fiber can deliver a regional, climate-positive alternative to peat that strengthens forest health, supports local economies, and meets growing demand from the horticulture sector.

Credits

We recognize and celebrate the innovation and leadership of Plumas Wood Fiber for advancing a practical, circular solution that converts excess forest biomass into a high-quality, peat-free horticultural substrate while strengthening regional wildfire resilience. We also extend our deep gratitude to [CALFIRE's Business and Workforce Development Program](#) for funding the Forest Business Alliance's work with [Plumas Wood Fiber](#), [Real Good Soil](#), and other community forestry partners across California. Rural and Tribal communities engaged in the forest products sector can benefit from resources, such as the Forest Business Alliance [Guidebook](#), the [Forest Business Alliance website](#), and upcoming Forest Business Alliance [workshops](#).

Unless indicated, all photos in figures by Jeff Greef.

Washoe Tribe Elder, Share & Sale Firewood Program

Christopher Sacco

Takeaways

- **Culturally Grounded Solutions.** The Washoe Tribe's Elder, Share & Sale Firewood Program addresses energy insecurity through a culturally rooted, community-driven approach that prioritizes elder care and traditional lifeways.
- **Integrated Environmental & Social Impact.** By sourcing firewood from forest thinning and restoration projects, the Program reduces wildfire risk and helps restore landscapes.
- **Innovative Cross-Subsidy Model.** A three-part structure—free firewood for elders, low-cost wood for disadvantaged households, and commercial sales—ensures equitable energy access and generates revenue to sustain operations.
- **Economic Development and Workforce Training.** The Program creates jobs for tribal members, and supports year-round training and employment, building a pipeline of future stewards.
- **Replicable & Scalable Solution.** The Washoe Tribe's strategic investment of tribal resources, use of grant funds, and development of partnerships with restoration programs demonstrates a scalable approach for other tribal and rural communities.

Context

The Washoe Tribe of Nevada and California, a federally recognized tribe of approximately 1,400 members, stewards ancestral homelands spanning some 64,300 acres across Alpine County, California, and Carson City, Douglas and Washoe Counties, Nevada—centered around Lake Tahoe (Dá'aw). For over 10,000 years, the Washoe people have maintained a reciprocal relationship with their environment, using fire as a vital tool for forest management, plant regeneration, and cultural practices like pinenut harvesting from piñon trees. This deep ecological knowledge fostered a balanced, adaptive system of land stewardship.

This long-standing relationship was disrupted by the westward expansion of European-American settlers who brought with them a fundamentally different worldview—one that saw land not as a living partner, but as “wilderness” that needed to be tamed. The settler colonial system prioritized ownership, resource extraction, and human domination over the environment. As part of this shift, settlers actively suppressed indigenous fire practices, viewing fire as inherently destructive rather than regenerative.

Fire suppression soon became a cornerstone of settler (and eventually Federal) land management policy, disrupting ecological cycles the Washoe and other tribes had long maintained. Forests once managed through cultural burning became overgrown, increasing the risk of catastrophic wildfires. These policies, combined with a focus on private property and extractive industries, directly clashed with indigenous approaches rooted in generational knowledge and shaped by millennia of adaptation to local ecosystems.

The consequences of these policies are visible today. The suppression of cultural burns has led to dense, unhealthy forests vulnerable to megafires, such as the 2021 Tamarack Fire, which burned 68,637 acres—including Washoe public domain allotments.¹ More recently, the June 2025 Conner Fire ignited near the Tribe's headquarters in Gardnerville, Nevada, scorching over 17,000 acres of the culturally and economically significant Pinenut Mountains. Although federal fire policy began to shift in the 1960s and incorporated prescribed burning by the 1990s and 2000s, fire suppression remains dominant in many areas, continuing to threaten both ecosystems and communities.

These environmental changes have direct impacts on the daily lives of Washoe people. Today, about 70% of tribal members heat their homes with firewood at least some of the time, and 33% rely solely on firewood.² For many elders—especially those living in rural or under-resourced homes in the Sierra Nevada region—wood is not just a necessity for survival during harsh winters, but also a connection to traditional lifeways and self-sufficiency.

¹Public domain allotments (PDAs) are parcels of U.S. federal public land allocated to individual Native Americans under policies like the Indian Homestead Act (1875) and Dawes Act (1887). These allotments were typically granted to Indians who lived off-reservation or belonged to tribes without established reservations. PDAs represent a form of individual Indian land tenure that is detached from tribal sovereignty. While tribes may not directly govern these lands, the allotments can affect tribal members' inheritance, water rights, and resource access. PDAs are managed by the Bureau of Indian Affairs if they are held in trust by the U.S. Government.

In response to these overlapping ecological and human challenges, the Washoe Tribe has launched initiatives to protect the land and its people. These include longstanding wildfire restoration and resilience, river and meadow restoration projects managed by the Washoe Environmental Protection Department, as well as the complementary Washoe Tribe Roads Department's Elder, Share & Sale Firewood Program. By restoring forest lands and use of beneficial fire and ensuring safe, equitable access to heating resources, the Tribe is working to reclaim stewardship practices disrupted by centuries of colonial policy—reviving a more sustainable relationship between land and community.



Figure 1: Jacob Vann, Elder, Share & Sale Program crewmember, at Washoe Wood Yard. Photo by Kaleb Roedel, Mountain West News Bureau.

Firewood program

The Washoe Elder, Share & Sale Firewood Program combines environmental stewardship and cultural responsibility with economic sustainability to meet the heating needs of tribal elders and other tribe members (Figure 1).

Overview

The “Elder” component functions much like a food bank, but instead of food, it delivers at least three cords of firewood—valued at roughly \$900—to each Washoe elder at no cost, helping them stay warm through harsh Sierra Nevada winters.³ The “Share” component provides firewood, at cost, to nearby tribes and disadvantaged Washoe members not eligible for the elder program. Finally, the “Sale” aspect, introduced in 2024, refers to the program’s cross-subsidy model, where surplus firewood—beyond what is needed for Elder and Share recipients—is sold at local gas stations, grocery stores, and campground supply shops. For every cord sold, the program funds the processing and delivery of another cord to an elder.

By linking commerce with care, the program prioritizes elder well-being while maintaining financial sustainability. Furthermore, it provides employment opportunities for tribe members, sustains forest stewardship and wood-burning practices that connect both youth and elders to their heritage and nature, and contributes to

²Cara Tabachnick and Alicia Hastey. 2025. “How wood gathered from federal forests heats Native American homes—and can help reduce wildfires.” CBS News. March 6.

³Kaleb Roedel. 2025. “How the Washoe Tribe built a business to sustain a firewood bank for their elders.” KUNR Public Radio. May 19.

environmental stewardship by sourcing wood from thinning, fire salvage, and habitat restoration projects in the region.

Evolution

The Program emerged out of a 2008 community-driven, volunteer effort where tribe members would gather and split firewood from local forests and donate bundles to elders. However, this informal model could not meet the demand—limited volunteers and an inconsistent supply of fuel wood meant that not all elders received sufficient firewood each winter. A major turning point came in 2021, when members volunteered to take logs damaged by the Tamarack Fire from the USFS. The Tribe invested its own resources to buy trucks and flatbed trailers and hire a part-time, seasonal crew. It also received a federal grant under the American Rescue Plan Act to purchase an excavator to unload trucks and to safely deck and manage the large logs, as well as chainsaws, personal protective equipment, and other firewood processing tools and equipment.

By 2022, the Elder component was providing three to four cords of firewood to over 160 Washoe Tribe Elders, and the Share component was supporting another 140+ Elders from other tribes who were struggling to afford the growing cost of firewood.

While the Washoe had made significant investments and secured grant funding, the Program still struggled to meet growing demand and many requests for firewood were going unfilled. The Program relied on six part-time, seasonal employees (all over the age of 40) and a combination of equipment—some never intended for commercial-scale production and others deteriorated from heavy, continuous use—leading to frequent breakdowns and a decline in production capacity despite the team's best efforts.

In response to these challenges, the Washoe Tribe sought grant funding to replace outdated equipment and adequately staff the program. In 2023, the Tribe secured a \$1 million grant from the USFS, enabling it to transition its six-person, part-time, seasonal crew to full-time, year-round positions. The grant also funded the purchase of industrial-grade wood processing equipment, including a gas-powered firewood processor, a commercial log splitter with a lift and conveyor system, a skid-steer loader with log grapple and stump buster attachments, a dump bed log trailer with grapple, a telehandler forklift, and an electric wrapper (Figure 2).



Figure 2: Field crew member operating electric wrapper at Washoe Wood Yard. Photo by Scott Burdette, USFS.

The firewood processor and industrial splitters allow for rapid processing of logs of all diameters (Figure 3). The skid-steer loader helps maintain a steady workflow for the wood processor, while the telehandler forklift speeds up the loading of firewood bags onto trailers and their efficient delivery to elders' homes (Figure 4). The electric wrapper bundles and wraps the firewood, making it easier to handle and deliver individual bundles as needed.



Figure 3: Field crew member operating industrial wood splitter at Washoe Wood Yard. Photo by Scott Burdette, USFS.

At current operating status, 883 CCF of raw materials per year are procured and processed into over 1,000 cords annually. Raw materials include sub-merchantable and non-merchantable small diameter logs (generally <12" DBH) – lodgepole, Jeffrey, and ponderosa pine with minor amounts of red and white fir. With the addition of the Sale component in June 2024, the Tribe has been able to sell enough wood to partially cover the field crew's salaries. The Elder, Share & Sale Firewood Program currently employs six Tribe members, yet because the Program is not yet self-sustaining year-round, crew members also support the USFS Meek Meadows Restoration Project and the Lake Tahoe Basin Management Unit Good Neighbor Authority Thinning Project during summer months.



Figure 4: Field crew member operating telehandler at Washoe Wood Yard. Photo by Scott Burdette, USFS.

Growing & sustaining

In May 2025, the Washoe Tribe was awarded a \$368,020 grant from the USFS to purchase a self-loading log truck, firewood trailer, log loader forks, and an additional gas-powered firewood processor. This funding will enable the Tribe to double its processing capacity, increasing its total utilization to 1,766 CCF per year—approximately 2,000 cords. This expansion will directly benefit Washoe Tribe elders across four non-contiguous reservations: the Carson Colony, Dresslerville Community, and Stewart Community in western Nevada, and the Woodfords Hung A Lel Ti Community in rural, eastern California.

To ensure long-term sustainability and support the scaling of the program, the grant also includes funds for a regional firewood business and market study. This study will assess additional demand for both bulk (cordwood) and bundled firewood, assist with outreach, and provide a financial/business plan to assist with cash flow analysis, sales forecasting, marketing strategies, and inventory management, all aimed at creating a financial model that supports informed decision-making.

The Tribe has identified a key gap in the need for a full-time program manager to handle contracting, fulfill firewood orders for elders, manage share and sale firewood, and oversee wood crew operations. Currently, this responsibility is being covered by the manager of another Tribal department. The Tribe is actively taking steps to address this gap, and once the marketing challenges are resolved, the program will be able to sustain a larger crew year-round, further increasing capacity and ensuring long-term sustainability.

The Elder & Share Firewood Program is not just about keeping homes warm. It also plays a vital role in forest management (Figure 5). Firewood is harvested from overgrown or burn-scarred landscapes—areas where fuel accumulation poses a significant wildfire risk. The Program procures 98% of its feedstock from vegetation thinning, fire salvage, and habitat restoration projects on federal lands, including the Lake Tahoe Basin Management Unit, Eldorado National Forest, Tahoe National Forest, and Humboldt-Toiyabe National Forest. Scaled up to 2,000 cords per year, the Program will contribute to restoring and reducing wildfire risk on 160 acres annually.



Figure 5: FBA/CAL FIRE-supported tour of Washoe Wood Yard. Photo by Regine Miller, FBA.

Finally, to support long-term stewardship of Washoe ancestral lands and develop the next generation of land managers, the Tribe's Woodfords Hung A Lel Ti Community launched the Washoe Tribe Youth Conservation Corps in 2023, with a \$3 million grant from the California Conservation Corps' Tribal Nature-Based Solutions Program. The Corps' restoration and workforce training efforts align closely with the Elder, Share & Sale Program's forest-health goals and serve as a pipeline for culturally grounded, skilled workers. As the Corps expands its operations, this program reinforces the Tribe's long-term vision of intergenerational stewardship, ecological resilience, and program sustainability.

Lessons learned

The Washoe Tribe's Elder & Share Firewood Program offers a powerful example of how tribal communities can combine cultural values, environmental stewardship, and economic innovation to meet community needs. Built on a foundation of resilience and adaptability, the Program has grown from an ad hoc, volunteer effort into a year-round operation that supports elder well-being, forest restoration, and workforce development—providing key lessons for others seeking to launch or expand similar initiatives:

1. Start-up & infrastructure

- Secure start-up funding (e.g., through grants or financing) to purchase essential equipment.
- Assess local infrastructure and identify regional limitations (e.g., lack of equipment to transport logs and the absence of nearby sawmills⁴) early in the planning process.
- Invest in durable, portable processing equipment to allow for on-site firewood manufacturing, including in remote or post-fire areas, as well as mobile equipment (e.g., trucks, excavator, skid steer), trailers, and bulk firewood bags or bins.

1. Operational Sustainability

- Develop a self-financing business model (e.g., cross-subsidizing elder deliveries by selling firewood commercially).
- Plan for equipment maintenance and replacement, anticipating wear and high utilization.
- Build a future pipeline of skilled workers.
- Build partnerships with federal agencies, nonprofits, local forestry contractors, and restoration programs to create stable supply chains.

1. Collaboration & Advocacy

- Engage with platforms, such as the Forest Business Alliance's peer learning network, and participate in site visits, to learn and share expertise.
- Advocate for policy changes that improve tribal access to forest lands and reduce restrictions on wood marketing and transportation.
- Push for increased state and federal investment, especially to close the funding gap in tribal forestry programs and wood infrastructure.

Leading by example

The Washoe Tribe's Program offers a compelling model for other Indigenous and rural communities across the West. It demonstrates that when Indigenous knowledge, community innovation, and partner support come together, the results can be transformative—not just for the land, but for the people who call it home. The Tribe's experience also underscores the long-term cultural and ecological damage caused by fire suppression policies. The 2025 Conner Fire, which destroyed tribal homes and vital infrastructure, devastated generations-old pine nut groves—key to Washoe cultural traditions and food sovereignty. Some of these trees take over 60 years to regenerate, and elders recall harvest areas that once sustained families, now lost to mismanagement or wildfire. This loss highlights the urgency of restoring Indigenous fire stewardship—not only as a tool for healthier landscapes, but to sustain cultural identity and community self-reliance. The Elder, Share & Sale Firewood Program is more than a heating solution. It is a working example of tribal self-determination in action—linking economic resilience, cultural continuity, and land stewardship. With the right tools, support, and leadership, other communities can build similarly grounded, self-sustaining programs that restore balance to both people and place.

Credits

We recognize and celebrate the Washoe Tribe of Nevada and California—their leaders, crew members, and entire community—for their vision and dedication to make the Elder, Share & Sale Firewood Program a reality. Their unwavering commitment to cultural continuity, community care, and ecological restoration lies at the heart of the Program's success and stands as a powerful example for other communities.

We also extend our sincere gratitude to [CALFIRE](#) for funding Forest Business Alliance's work with tribal and community forestry partners across California. This support enabled us to provide crucial technical assistance that helped the Washoe Tribe secure a major 2025 USFS grant—an investment that has significantly increased the Tribe's capacity to meet community heating needs while promoting sustainable forest stewardship. FBA is proud to have contributed, in a small way, to supporting the Washoe Tribe in the development and expansion of this innovative program.

⁴“Trump timber policies could boost tribal revenues, but infrastructure gaps remain”. Tribal Business News. August 10, 2025.

Finally, we thank the many restoration partners and feedstock suppliers—including national forests, conservation organizations, and local contractors—whose collaboration ensures a consistent, sustainable supply of wood from forest thinning, fire recovery, and habitat restoration efforts. Their shared dedication to forest health and tribal self-determination has made this important work possible.